

Emerging cooperativity between Oct4 and Sox2 governs the pluripotency network in mouse early embryos

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Abstract

During the first lineage segregation, mammalian embryos generate the inner cell mass (ICM) and trophoctoderm (TE). ICM gives rise to the epiblast (EPI) that forms all cell types of the body, an ability referred to as pluripotency. The molecular mechanisms that induce pluripotency in embryos remain incompletely elucidated. Using knockout (KO) mouse models in conjunction with low-input ATAC-seq and RNA-seq, we found that Oct4 and Sox2 gradually come into play in the early ICM, coinciding with the initiation of Sox2 expression. Oct4 and Sox2 directly activate the pluripotency-related genes through the corresponding OCT-SOX enhancers in the early ICM. Furthermore, we observed a substantial reorganization of chromatin landscape and transcriptome from the morula to the early ICM stages, which was partially driven by Oct4 and Sox2, highlighting their pivotal role in promoting the developmental trajectory towards the ICM. Our study provides new insights into the establishment of the pluripotency network in mouse preimplantation embryos.

eLife Assessment

This work presents a **valuable** finding on how the interplay between transcription factors SOX2 and OCT4 establishes the pluripotency network in early mouse embryos. Despite the high quality of the data, the evidence supporting the claims of the authors is currently **incomplete** and would benefit from more omics analysis such as H3K4me1 and H3K27ac CUT&Tag. The work will be of interest to biologists working on embryonic development.

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Introduction

Following fertilization, mouse embryos undergo a series of cell divisions and differentiation in the first few days. Zygotes and 2-cell blastomeres are totipotent and can form both embryonic and extra-embryonic supporting tissues. After several rounds of divisions, embryos undergo compaction at the morula stage and specifies the trophectoderm (TE) and inner cell mass (ICM). The ICM further segregates into the primitive endoderm (PE) and the epiblast (EPI) in the subsequent blastocyst stage. All cell types of the fetus can develop from the EPI, a feature known as pluripotency. During the establishment of pluripotency, dramatic molecular changes occur such as shifts in epigenetic modifications, chromatin accessibility and transcriptome (Deng *et al.*, 2014 [DOI](#); Guo *et al.*, 2010 [DOI](#); Wang *et al.*, 2018 [DOI](#); Wu *et al.*, 2016 [DOI](#); Zhang *et al.*, 2016 [DOI](#); Zhang *et al.*, 2018 [DOI](#)). The precise mechanisms involved in establishing the transient pluripotent state during development remain elusive.

Oct4 (also known as *Pou5f1*) and Sox2 are prominent transcription factors (TFs) that regulate early embryonic development. In mice, Oct4 is expressed in all the cells of the compacted morulae, while Sox2 is first expressed in the inside cell and identified as one of the earliest markers to distinguish the inner from the outer cells (Guo *et al.*, 2010 [DOI](#); Palmieri *et al.*, 1994 [DOI](#); White *et al.*, 2016 [DOI](#)). Both factors are confined to the pluripotent EPI at the late blastocyst stage (Avilion *et al.*, 2003 [DOI](#); Palmieri *et al.*, 1994 [DOI](#)). Oct4 and Sox2 knockout (KO) leads to embryonic lethality around embryonic day (E) 4.5 and E6.0, respectively (Avilion *et al.*, 2003 [DOI](#); Nichols *et al.*, 1998 [DOI](#)). Although Oct4- or Sox2-KO embryos develop apparently normal EPI, they fail to give rise to embryonic stem cells (ESCs) (Avilion *et al.*, 2003 [DOI](#); Nichols *et al.*, 1998 [DOI](#)) and Oct4- KO embryos cannot contribute to embryonic tissues in chimera assays (Wu *et al.*, 2013 [DOI](#)), suggesting that Oct4 and Sox2 are required for the EPI to acquire or maintain pluripotency. Surprisingly, Oct4- and Sox2-KO mouse embryos still express many pluripotency markers, such as Nanog, *Klf4* and *Zfp42* (*Rex1*) (Avilion *et al.*, 2003 [DOI](#); Frum *et al.*, 2013 [DOI](#); Le Bin *et al.*, 2014 [DOI](#); Stirparo *et al.*, 2021 [DOI](#); Wicklow *et al.*, 2014 [DOI](#); Wu *et al.*, 2013 [DOI](#)). The controversy between the loss of pluripotency and the maintenance of the pluripotency-related genes in Oct4- and Sox2- KO EPI prompted us to investigate the role of Oct4 and Sox2 in early mouse embryos.

Recent studies suggested that Oct4 and Sox2 may function before the blastocyst stage. The long-lived presence of exogenous Oct4 and Sox2 proteins on DNA was observed in some of the 4-cell blastomeres, that tended toward the ICM rather than the TE (Plachta *et al.*, 2011 [DOI](#); White *et al.*, 2016 [DOI](#)), suggesting that they may play a role in the initial lineage segregation between ICM and TE. Oct4 is also thought to contribute to the dramatic gain of chromatin accessibility at the 8-cell stage (Lu *et al.*, 2016 [DOI](#)). Unlike in mouse embryos, the loss of OCT4 expression in human embryos compromises blastocyst development, leading to the downregulation of *NANOG* in the ICM and *CDX2* in the TE (Fogarty *et al.*, 2017 [DOI](#)). Oct4 controls zygotic genome activation (ZGA) in zebrafish and human, but not in mice (Gao *et al.*, 2018 [DOI](#); Leichsenring *et al.*, 2013 [DOI](#)). Thus, when and how Oct4 and Sox2 function in the early embryogenesis varies from species to species.

Oct4 and Sox2 play preeminent roles in maintaining the pluripotency-related transcriptional network in ESCs (Masui *et al.*, 2007 [DOI](#); Niwa *et al.*, 2000 [DOI](#)). However, it remains unclear how these two TFs regulate pluripotency-related genes in embryos. The obvious obstacles are the heterogeneity and the scarcity of the preimplantation embryos. Unlike ESCs, which comprise a homogenous cell population, individual cells within embryos gradually differentiate into three cell types, EPI, PE and TE. This inherent cellular heterogeneity within embryos, coupled with the temporal variability in development across different embryos, poses a significant challenge when investigating the molecular processes using conventional molecular techniques. For instance, using the entire blastocyst or ICM for qPCR and bulk RNA-seq analysis would inevitably lead to a dilution and blur of the molecular processes within EPI (Frum *et al.*, 2013 [DOI](#); Wu *et al.*, 2013 [DOI](#)).

Although single-cell genomics is powerful to dissect cellular heterogeneity, it suffers from high dropouts, which may lead to inaccuracies in genomic profiles and cell identities (Minow *et al*, 2023 [↗](#)). Due to the scarcity of embryos, it is unfeasible to compensate for the inaccuracy by increasing the number of cells. For example, single-cell RNA-seq (scRNA-seq) still faces challenges in accurately distinguishing between ICM and TE progenitors in compacted morula, as well as between EPI and PE progenitors in the early ICM (Deng *et al.*, 2014 [↗](#); Li *et al*, 2023 [↗](#); Yanagida *et al*, 2022 [↗](#)).

MEK inhibitor (MEKi) promotes the ground state of ESCs by suppressing the PE-inducing ERK signaling pathway (Ying *et al*, 2008 [↗](#)). In embryos, MEKi has been shown to activate Nanog in ICMs, effectively suppressing the development of PE and directing the entire ICM towards a pluripotent epiblast fate (Nichols *et al*, 2009 [↗](#)). The naïve pluripotency and readiness for differentiation of MEKi-treated EPI has been confirmed by its contribution to chimaeras with germline transmission. MEKi may thus facilitate the study of molecular events within the EPI cells by reducing the heterogeneity in ICMs.

In this study, we analyzed the transcriptome and global chromatin accessibility in the *Oct4*- or *Sox2*-KO mouse embryos using low-input RNA-seq and ATAC-seq. The effects of *Oct4* KO and *Sox2* KO are relatively small in morulae, but become evident in ICMs at the blastocyst stage. *Oct4* and *Sox2* mainly activate pluripotency-related genes cooperatively through the enhancers containing the composite OCT-SOX motifs. We observed a substantial reorganization of open chromatin regions and transcriptome in the early ICM, which was disturbed in absence of either *Oct4* or *Sox2*. These results indicate the critical roles of *Oct4* and *Sox2* in establishing the pluripotency network during early mouse development.

Results

Generation of maternal-zygotic KO embryos

To investigate the role of *Oct4* and *Sox2* in the development of preimplantation embryos, we produced four transgenic mouse lines: *Oct4* KO labeled with mKO2 (monomeric kusabira- orange 2, *Oct4*^{mKO2}), *Sox2* KO labeled with EGFP (*Sox2*^{EGFP}), floxed *Oct4* (*Oct4*^{flox}) and floxed *Sox2* (*Sox2*^{flox}) (Figure 1A [↗](#), 1B [↗](#); Materials and methods). Although the expression of mKO2 and EGFP is driven by the constitutive mouse PGK promoter, we observed that the expression of these reporters is weak until the 8-cell and blastocyst stages, respectively, when the strong zygotic expression of *Oct4* and *Sox2* is observed.

Oct4^{+/mKO2} and *Sox2*^{+/EGFP} heterozygous male mice were mated with *Oct4*^{flox/flox}; *ZP3-Cre* and *Sox2*^{flox/flox}; *ZP3-Cre* maternal KO female mice, respectively, to generate fluorescently labeled maternal-zygotic KO and unlabeled maternal KO (herein called control, Ctrl) embryos (Figure 1C [↗](#), 1D [↗](#)). These reporter systems allow prospective identification of KO embryos using fluorescent microscopy (Figure 1-figure supplement 1 [↗](#)), so we were able to pool embryos with the same genotype. Immunostaining results confirmed that the mKO2⁺ and EGFP⁺ embryos lost *Oct4* and *Sox2* proteins, respectively (Figure 1E [↗](#), 1F [↗](#)).

Oct4 and *Sox2* regulate the chromatin landscape and transcriptome in the ICM

To assess the genome-wide molecular impact of the loss of *Oct4* and *Sox2*, we performed low-input ATAC-seq and RNA-seq using early (E2.75) and late (E3.25) compacted morulae, as well as early (E3.75) and late (E4.5) ICMs (Materials and methods). For ATAC-seq, we pooled embryos based on our reporter systems because single-embryo samples yielded sparse signals and too few peaks (data not shown). We opted to collect single morulae or ICMs for RNA-seq, as this approach

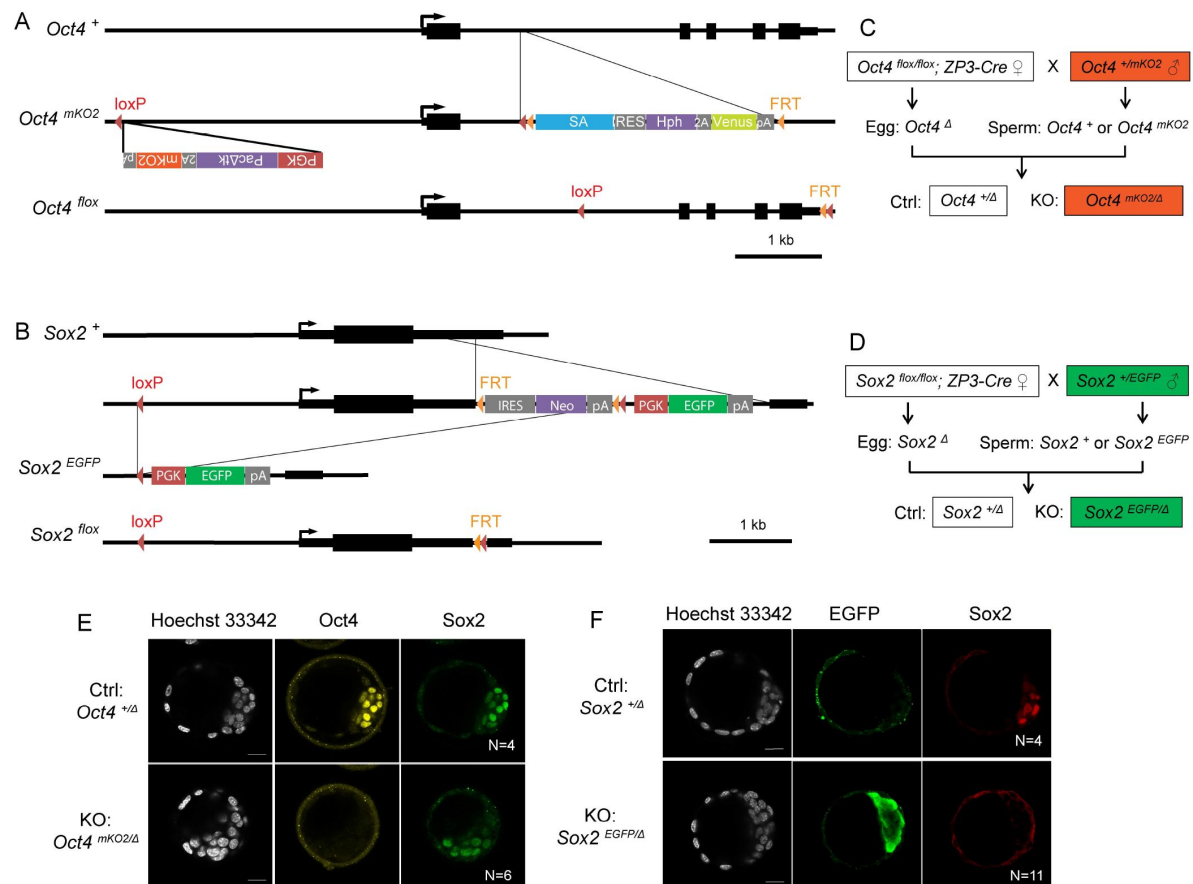


Figure 1.

Generation of maternal-zygotic KO embryos.

A. Schemes of mKO2-labeled *Oct4* KO (*Oct4*^{mKO2}) and *Oct4*^{flox} alleles. **B.** Schemes of EGFP-labeled *Sox2* KO (*Sox2*^{EGFP}) and *Sox2*^{flox} alleles. **C.** Mating strategy for *Oct4* Ctrl and KO embryos. **D.** Mating strategy for *Sox2* Ctrl and KO embryos. **E.** Immunostaining with embryos separated based on the mKO2 fluorescence. **F.** Immunostaining with embryos separated based on the EGFP fluorescence. N, the number of embryos. Scalebar, 20 μ m.

enabled us to account for embryo-to-embryo variability and detect transcripts with greater sensitivity compared to scRNA-seq (Boroviak *et al.*, 2015 [\[1\]](#)). Given that the ICM consists of the progenitors of EPI and PE cells, we treated the embryos with the MEKi (PD0325901) from E2.5 to suppress PE development and specify the entire ICM to the EPI (Nichols *et al.*, 2009 [\[2\]](#)) (**Figure 1-figure supplement 2** [\[3\]](#)).

Principal component analysis (PCA) of the ATAC-seq data reveals a clear clustering of samples based on the stages and genotypes (**Figure 2A** [\[4\]](#)). The *Oct4*-KO early and late morulae clustered close to their Ctrl counterparts, while the *Oct4*- and *Sox2*-KO early and late ICM samples clustered separately from their Ctrl. In total, 152,393 peaks were identified across different stages and genotypes. Of these, 14,016 (9.2%) and 6,637 (4.4%) showed significant decreases, while 11,884 (7.8%) and 3,660 (2.4%) exhibited significant increases in *Oct4*- and *Sox2*-KO late ICM, respectively (**Figure 2B** [\[5\]](#); Data supplement 1). The number of decreased peaks exceeded that of increased peaks across all the KO samples. K-means clustering of all the 33,520 differential peaks shows that many peaks, which exhibited reduced accessibility in the KO ICMs, gained accessibility from the Ctrl morula to the Ctrl ICM (**Figure 2C** [\[6\]](#), clusters 1, 3, 8, 10, 11, 13 and 14). Notably, the peaks activated by *Oct4* but not by *Sox2* in the ICM tended to be already open at the morula stage (**Figure 2B** [\[5\]](#), clusters 1 and 11), whereas those dependent on both *Oct4* and *Sox2* became open in the ICM (**Figure 2B** [\[5\]](#), clusters 3, 8 and 14). Over 96.8% of the significantly changed peaks were identified as putative enhancers located distally to the transcription start sites (TSSs) (**Figure 2-figure supplement 3A** [\[7\]](#)).

To validate the authenticity of the *Sox2*-regulated downstream ATAC-seq peaks identified in this study, we compared the data to a recent study that did not employ MEKi treatment (Data ref: Li *et al.*, 2023 [\[8\]](#)) (Li *et al.*, 2023 [\[8\]](#)). The trends observed in our *Sox2*-KO late ICMs, including decreased, increased, and unchanged peaks, remained consistent in the absence of MEK inhibitor (**Figure 2-figure supplement 3B** [\[9\]](#)). Remarkably, the decreased ATAC-seq peaks also showed higher enrichment with *Sox2* CUT&RUN signals than the increased or unchanged peaks (**Figure 2-figure supplement 3C** [\[10\]](#)). Analysis of published ChIP-seq data in ESCs (Data ref: Marson *et al.*, 2008 [\[11\]](#); Whyte *et al.*, 2012 [\[12\]](#)) (Marson *et al.*, 2008 [\[11\]](#); Whyte *et al.*, 2012 [\[12\]](#)) shows that the enhancers decreasing in the *Oct4*- or *Sox2*-KO late ICMs were frequently bound by *Oct4*, *Sox2* and *Nanog*, and enriched with the active enhancer mark H3K27ac (**Figure 2-figure supplement 3D** [\[13\]](#)). The *Sox2*-dependent peaks around the *Sap30* gene are shown as examples (**Figure 2-figure supplement 3E** [\[14\]](#)). These data suggest that the decreased peaks identified in our ATAC-seq could be the direct targets of *Oct4* and *Sox2* in the ICM.

To assess the impact of *Oct4* and *Sox2* deletion on the transcriptome, we performed RNA-seq on individual morulae or ICMs. Initially, we examined the consequences of maternal KO and zygotic heterozygous KO. PCA did not reveal significant distinctions between maternal flox and maternal KO, nor between WT/flox and flox/KO samples (**Figure 2-figure supplement 4A** [\[15\]](#)). Furthermore, mice with heterozygous KO of *Oct4* or *Sox2* develop normally and exhibit fertility (Nichols *et al.*, 1998 [\[16\]](#)). Therefore, we employed maternal-KO zygotic-WT/KO as our Ctrl group. In morulae, *Oct4* KO had only a minor effect on the transcriptome (**Figure 2D** [\[17\]](#)). We observed a progressively profound impact of *Oct4* KO on the transcriptome from the morula to the late ICM, mirroring patterns at the chromatin level (**Figure 2A** [\[4\]](#)–**2D** [\[17\]](#)). As *Sox2* is only expressed at very low levels until the blastocyst stage anyway, the transcriptome in *Sox2*-KO morulae was not significantly affected. *Oct4* or *Sox2* KO altered the expression of 2,485 (15.8%) and 967 (6.1%) genes in the late ICM, respectively (**Figure 2D** [\[17\]](#); Data supplement 2). *Oct4* KO had a greater effect than *Sox2* KO on both the chromatin accessibility and the transcriptome (**Figure 2A** [\[4\]](#)–**2D** [\[17\]](#)), suggesting *Oct4* may play a more important role than *Sox2*. This is consistent with the earlier developmental arrest of *Oct4*-KO embryos at E4.5 compared to *Sox2*-KO embryos at E6.0. Down- and up-regulated genes identified in our *Oct4*- and *Sox2*- KO ICMs show enrichment with genes exhibiting down- and up-

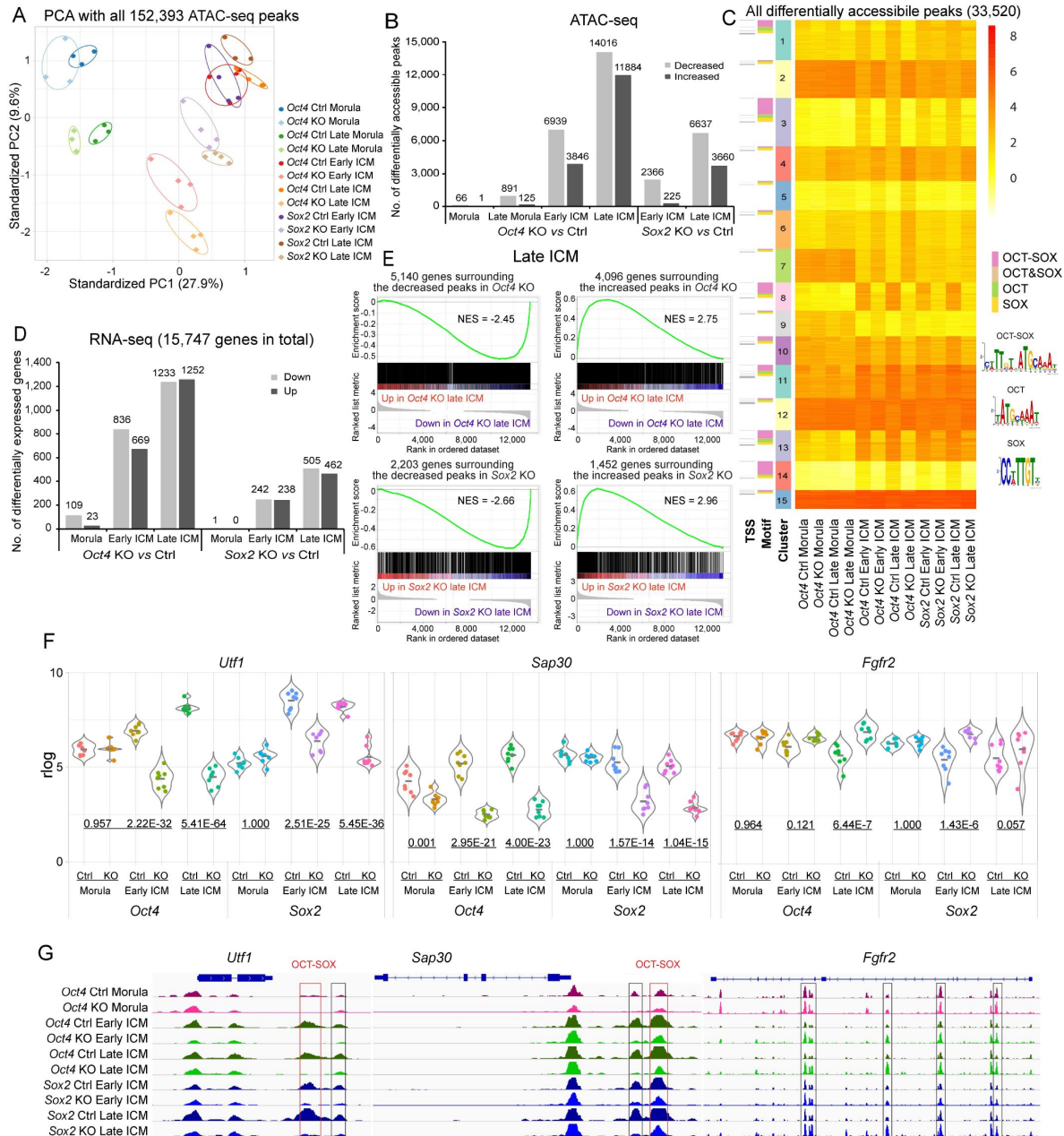


Figure 2.

Oct4 and Sox2 regulate chromatin landscape and transcriptome in the ICM.

A. PCA plot of all the identified ATAC-seq peaks. **B.** Number of differentially accessible ATAC-seq peaks in KO vs Ctrl samples. Cutoff, adjusted p-value <0.05. **C.** k-means clustering of all the significantly differential ATAC-seq peaks in KO vs Ctrl in **Figure 2B**. The heatmap is sorted by clusters, motifs and TSS. Peaks located within 100 bp from TSS were considered as TSS peaks. OCT-SOX, OCT or SOX motifs indicate that the peak contains the canonical OCT-SOX, OCT or SOX motif, respectively, while OCT&SOX motif indicates that separate OCT and SOX motifs were discovered in one peak. Cutoff, adjusted p-value <0.05. **D.** Number of differentially expressed genes in KO vs Ctrl samples. Cutoff, adjusted p-value <0.05 and log2 fold change ≥ 1 . **E.** GSEA shows the correlation between significantly changed ATAC-seq peaks and the transcription of genes located within 10 kb of the peaks in Oct4- or Sox2-KO late ICMs. NES, normalized enrichment score. **F.** Examples of down- and up-regulated genes. The underlined numbers represent the adjusted p values. **G.** The ATAC-seq profiles surrounding the genes in E. Boxes mark the differentially accessible peaks, and red boxes specifically mark those with the OCT-SOX motif.

regulation, respectively, in the studies without MEKi (Data ref: [Li et al., 2023](#); [Stirparo et al., 2021](#)) ([Li et al., 2023](#); [Stirparo et al., 2021](#)) (**Figure 2-figure supplement 4B-4D**), confirming that MEKi did not alter the molecular functions of Oct4 and Sox2 in the ICM.

Next, we explored whether changes in chromatin accessibility affected the transcription of surrounding genes. Gene set enrichment analysis (GSEA) shows that the genes close to the decreased peaks were frequently downregulated, whereas those near the increased peaks were upregulated in the *Oct4*- and *Sox2*-KO embryos (**Figure 2E**; **Figure 2-figure supplement 5**). For example, the expression of *Utf1*, a known target of Oct4 and Sox2, and its enhancer accessibility decreased in the *Oct4*- and *Sox2*-KO ICMs (**Figure 2F**, **2G**). On the other hand, the expression of *Fgfr2* and its enhancer accessibility increased in the *Oct4*- and *Sox2*-KO ICM. In addition, integration of the ATAC-seq and RNA-seq data allowed us to identify previously unknown targets of Oct4 and Sox2, such as *Sap30* and *Uhrf1*, which are essential for somatic cell reprogramming and embryonic development (**Figure 2F**, **2G**) ([Cao et al, 2019](#); [Li et al, 2017](#); [Maenohara et al, 2017](#)).

Taken together, the data presented so far indicate that Oct4 and Sox2 play a crucial role in shaping the chromatin landscape and transcriptome in the ICM. Consequently, we observed apparently normal development of the *Oct4*- and *Sox2*-KO embryos up to the blastocyst stage.

Oct4 and Sox2 activate the pluripotency network in the ICM

To further explore the role of Oct4 and Sox2, we wanted to find out which genes they might influence in the ICM. As expected, many of the differential ATAC-seq peaks were consistently changed in *Oct4*- and *Sox2*-KO late ICMs (**Figure 3A**, Data supplement 1). However, there are a number of peaks exclusively changed in either *Oct4*- or *Sox2*-KO ICMs. The genes around the consistently decreased peaks were enriched for terms related to pluripotency and preimplantation embryonic development, such as cellular response to LIF signaling and stem cell population maintenance (**Figure 3B**, upper panel). The consistently elevated peaks preferentially located near the genes related to extraembryonic lineages and organogenesis, such as embryonic placenta, extraembryonic trophoblast, muscle and neural tube (**Figure 3B**, lower panel). Interestingly, the genes near the peaks which decreased only in the *Oct4*-KO but not in the *Sox2*-KO ICM were enriched with terms of LIF signaling and blastocyst formation (**Figure 3-figure supplement 6**).

At the transcriptional level, many known pluripotency genes, such as *Klf2*, *Etv5*, *Prdm14* and *Pecam1*, were downregulated in the *Oct4*- and *Sox2*-KO ICM (**Figure 3C**). In contrast, *Gata3*, *Cdx2* and *Eomes*, which are important for TE development and differentiation ([Ralston et al, 2010](#); [Russ et al, 2000](#); [Strumpf et al, 2005](#)), were upregulated in the *Oct4*- and *Sox2*-KO ICM.

Interestingly, we observed that *Nanog*, *Esrrb* and *Klf4* were significantly downregulated in the *Oct4*-KO ICM, whereas they were not or only slightly downregulated in the *Sox2*-KO ICM (**Figure 3C**, **3D**). Downregulation of *Pecam1* was confirmed at the protein level (**Figure 3E**). Chromatin accessibility at its putative enhancers also decreased accordingly (**Figure 3F**). Oct4 and Sox2 activated the components of several epigenetic modifiers, such as *Ezh2* (PRC2 H3K27me2/3 methyltransferase) and *Sap30* (a component of mSin3A histone deacetylase complex) (**Figure 2F**, **2G**; Data supplement 2), suggesting their potential contribution to establishing the ICM-specific epigenetic status through regulation of the epigenetic modifiers.

Taken together, the above data show that Oct4 and Sox2 regulate a large number of TFs, epigenetic factors and signaling pathways in the ICM.

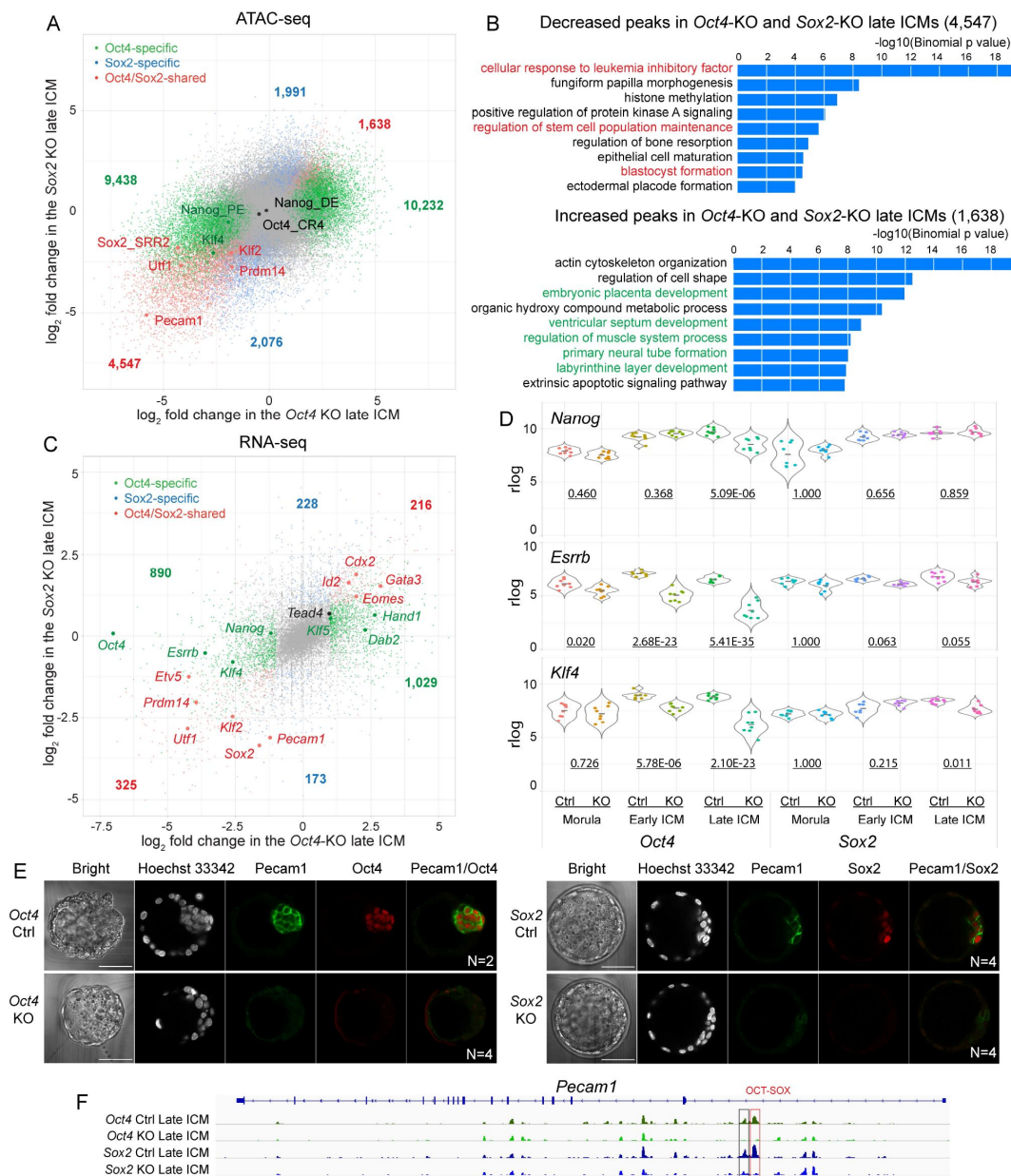


Figure 3.

Oct4 and Sox2 activate EPI-specific genes and suppress TE-specific genes in ICM.

A. A scatter plot shows the log₂ fold change of ATAC-seq signals in the Oct4-KO and Sox2-KO late ICM. Colored dots represent the significantly changed peaks. Cutoff, adjusted p-value <0.05. **B.** GREAT ontology enrichment analysis of the significantly changed peaks shared in the Oct4-KO and Sox2-KO late ICM. Red terms are related to the pluripotency and preimplantation embryonic development, and green ones are related to the development of embryonic and extraembryonic lineages in the post-implantation embryos. **C.** A scatter plot shows the log₂ fold changes of RNA-seq signals in the Oct4-KO and Sox2-KO late ICM. Cutoff, adjusted p-value <0.05 and log₂ fold change ≥1. **D.** Violin plots of RNA-seq rlog values for *Nanog*, *Esrrb* and *Klf4* in the embryos. The underlined numbers represent the adjusted p values. **E.** Immunostaining of Pecam1 in the late blastocysts (E2.5+2 days). Scalebar, 50 μm. **F.** The accessibility of putative enhancers around *Pecam1* in ICMs. Boxes mark the decreased peaks and red box marks the peak with the OCT-SOX motif.

Oct4 and Sox2 co-activate their targets through OCT-SOX enhancers

Although there is a general consensus on the cooperative binding of Oct4 and Sox2 to the OCT-SOX composite motif, the principle of the cooperation still remains controversial in different scenarios (Biddle *et al*, 2019 [↗](#); Chen *et al*, 2014 [↗](#); Friman *et al*, 2019 [↗](#); Li *et al*, 2019 [↗](#); Michael *et al*, 2020 [↗](#)). Therefore, we investigated how Oct4 and Sox2 regulate their target enhancers in the ICM. As expected, the OCT-SOX, OCT and SOX motifs were also among the most enriched motifs in the group of decreased enhancers in *Oct4*- or *Sox2*-KO ICMs (Figure 4A [↗](#)). The motifs of other known pluripotency-related TFs, such as Klf2/4 and Esrrb, were also enriched at the decreased enhancers. The elevated peaks were enriched for the GATA, TEAD, EOMES and KLF motifs, but not for the OCT-SOX, OCT or SOX motifs. Additionally, the gain of chromatin accessibility occurred at later stages compared to the loss of chromatin accessibility (Figure 2B [↗](#)). Thus, the increased peaks may not represent the direct targets of Oct4 or Sox2, but rather may be a secondary effect of the upregulated TE TFs, such as *Gata3*, *Tead4*, *Eomes* and *Klf5/6*, or the disruption of inhibitory interactions between Oct4/Sox2 and TE TFs (Niwa *et al*, 2005 [↗](#)) (Figure 3C [↗](#); Data supplement 2). As a control, the CTCF motif was not enriched in either the decreased or increased peaks. We confirmed the enrichment of those motifs at distal peaks and concluded that the motif analysis was not biased by the small number of the TSS peaks in the dataset (Figure 2-figure supplement 3A [↗](#), Figure 4-figure supplement 7 [↗](#)).

To investigate the effect of Oct4 and Sox2 on the chromatin accessibility, we focused on the peaks in the ICM. It is worth noting that the top decreased peaks in the *Oct4*- and *Sox2*-KO early ICM were most enriched with the OCT-SOX, but not the OCT or SOX motifs (Figure 4B [↗](#)), which suggests that Oct4 and Sox2 could maintain open chromatin to a greater extent at peaks containing the OCT-SOX motif (hereafter referred to as OCT-SOX peaks). The KLF motif was more enriched at the moderately decreased peaks, indicating a possible secondary effect of down-regulated *Klf2/4* (Figure 3C [↗](#), 3D [↗](#)); alternatively, the depletion of Oct4 and Sox2 might lead to a decrease in DNA binding or the activity of *Klf2/4* and other KLF TFs. In general, the accessibility of 8,993 OCT-SOX peaks decreased in both the *Oct4*- and *Sox2*-KO early and late ICM (Figure 4C [↗](#); Figure 4-figure supplement 8 [↗](#)). Furthermore, majority of the decreased OCT-SOX peaks were shared in *Oct4*- and *Sox2*-KO late ICM (Figure 4D [↗](#)), suggesting that both TFs participate in keeping these peaks open. The known OCT-SOX enhancers of *Dppa3* and *Klf4*, which were enriched with the binding of Oct4 and Sox2, are shown as examples (Figure 4E [↗](#)).

The above results suggest that the enhancers containing the OCT-SOX motif are primary targets of Oct4 and Sox2 in the ICM.

Oct4 and Sox2 promote the developmental trajectory towards ICM

In early embryos, individual blastomeres are initially totipotent and indistinguishable before segregating into the ICM and TE. We subsequently investigated factors that influence the cellular trajectory toward the pluripotent ICM. Around 50% of the peaks and genes exhibiting decreased trend in *Oct4*- or *Sox2*-KO early ICMs were found to be activated in the Ctrl ICM during the natural development (Figure 5A [↗](#)). In alignment with earlier studies (Guo *et al*, 2010 [↗](#); Wu *et al*, 2016 [↗](#)), a significant reorganization of open chromatin regions was observed upon the formation of the early ICM, resulting in the activation of 21,731 peaks (14.3%) (Figure 5B [↗](#)). The most prominently elevated peaks in the early ICM locate close to genes involved in cellular response to LIF, maintenance of stem cell population and blastocyst formation, in accordance with the developmental stage (Figure 5-figure supplement 9A [↗](#)). These peaks were enriched for OCT, SOX and OCT-SOX motifs (Figure 5-figure supplement 9B [↗](#)). However, when Oct4 and Sox2 were absent, the accessibility of the elevated peaks was decreased in the early ICM (Figure 5C [↗](#)). In alignment with the chromatin dynamics, the transcriptome also underwent substantial rearrangements in early ICMs, resulting in the upregulation of 1,115 genes (7.1%) (Figure 5D [↗](#)).

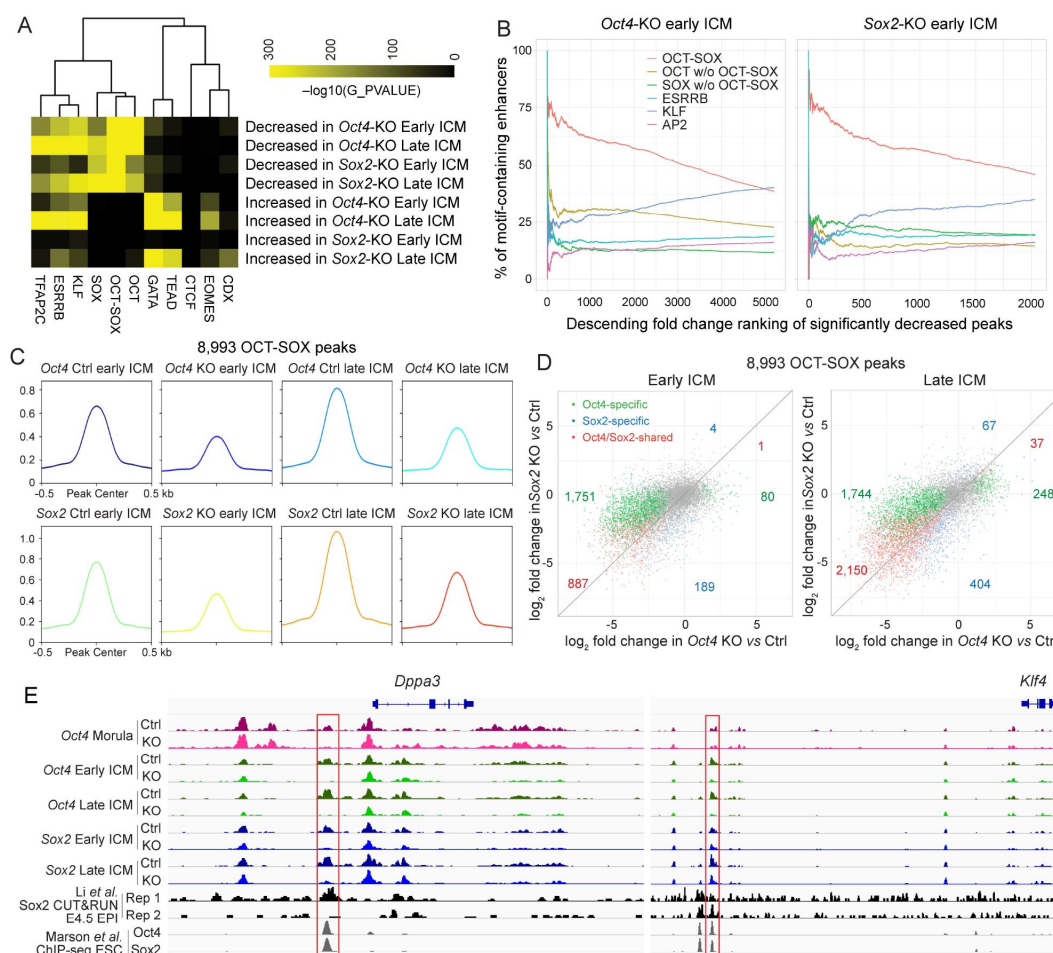


Figure 4.

Oct4 and Sox2 activate OCT-SOX enhancers cooperatively and independently in ICM.

A. Motif enrichment analysis of significantly changed ATAC-seq peaks in the *Oct4*-KO and *Sox2*-KO ICMs. **B.** Occurrence of the motifs in the ranked peaks. The decreased enhancers were ranked by the fold reduction. The cumulative percentages of peaks containing at least one sequence for a given motif are plotted against ranks of peaks. **C.** Average profiles of the 8,993 OCT-SOX peaks across all the Ctrl and KO ICM samples. **D.** Scatter plots show the $\log_2$ fold change of 8,993 OCT-SOX peaks in the *Oct4*- and *Sox2*-KO late ICMs. **E.** The profiles of ATAC-seq in early embryos, Sox2 CUT&RUN in E4.5 EPI (Li *et al.*, 2023) and Oct4 and Sox2 ChIP-seq in ESCs (Marson *et al.*, 2008) around the known OCT-SOX enhancers of *Klf4* and *Dppa3*. The red boxes mark the OCT-SOX enhancers.

These 1,115 genes were enriched with genes downregulated in the *Oct4*-KO and *Sox2*-KO early ICM (Figure 5E). These data indicate a critical role of Oct4 and Sox2 in initiating ICM-specific chromatin and transcriptional programs.

The network of naïve pluripotency is governed by a core regulatory circuit of TFs, including Oct4, Sox2, Nanog, Esrrb and Klf4 (Adachi *et al.*, 2018; Ng & Surani, 2011; Young, 2011). In particular, the activation of endogenous Sox2 represents a late-stage and deterministic event that triggers the fate towards pluripotency during reprogramming (Buganim *et al.*, 2012). In mouse embryos, all the aforementioned TFs except Sox2 are highly expressed in the morula (Figure 5-figure supplement 9C); however, their expression does not seem sufficient to induce the mature pluripotent state at this stage. Therefore, we hypothesized that Sox2, in collaboration with Oct4, may activate the pluripotency-related genes in the early ICM. To investigate this hypothesis, we subjected the 1,115 genes and their log₂foldchange in *Sox2*-KO and *Oct4*-KO early ICMs to GSEA Wikipathway analyses (Liao *et al.*, 2019). This analysis revealed an enriched downregulation of genes associated with PluriNetWork and ESC pluripotency pathways in *Sox2*-KO early ICMs (Figure 5F). In *Oct4*-KO early ICMs, although such enrichment was not observed, a significant downregulation of pluripotency-related genes was evident (Figure 5-figure supplement 9D, 9E). For example, in the early ICM, Oct4 and Sox2 activate *Utf1* and *Il6st* (also known as *gp130*, receptor of the LIF/STAT pathway) through the OCT-SOX enhancers (Figure 2F, 2G, 5H). Notably, the OCT-SOX enhancer of *Il6st* is enriched with Sox2 CUT&RUN signals. Additionally, Oct4 activated enzymes regulating the pyruvate metabolism (*Ldha*, *Me2*, *Pck2*, etc.) and glutathione metabolism (*Gstm1/2*, *Mgst2/3*, *Idh1*), consistent with previous studies highlighting Oct4's role in metabolism regulation (Frum *et al.*, 2013; Stirparo *et al.*, 2021). These findings indicate that Sox2 expression might function as a temporal regulator in the activation of OCT-SOX enhancers and a subset of pluripotency-related genes in the early ICM.

In summary, above data suggest that Oct4 and Sox2 play pivotal roles in promoting the ICM fate in early mouse embryos.

Discussion

Oct4 and Sox2 are key TFs for preimplantation embryonic development in several species (Daigneault *et al.*, 2018; Fogarty *et al.*, 2017; Gao *et al.*, 2022; Lee *et al.*, 2013; Leichsenring *et al.*, 2013). However, because of the paucity of embryonic cells, it remains a matter of debate as when they start to function and how they regulate the development of the EPI lineage. In this study, we explored the roles of Oct4 and Sox2 in mouse preimplantation embryos using transgenic mouse models.

Previous reports suggested that Oct4 and Sox2 could direct 4-cell blastomeres towards ICM fate at the first lineage segregation (Plachta *et al.*, 2011; White *et al.*, 2016). siRNA-mediated knockdown experiments showed that Oct4 facilitates chromatin opening already at the 8-cell stage (Lu *et al.*, 2016). Unexpectedly, our data showed that either *Oct4*-KO or *Sox2*-KO have minimal impact on both global chromatin accessibility and transcription until the early blastocyst stage (Figure 2A–2D). Oct4 is expressed at similar levels between morulae and ICMs; in contrast, Sox2 mRNA and protein levels are negligible in morulae, but are significantly upregulated in ICMs (Figure 5-figure supplement 9C) (Palmieri *et al.*, 1994; Wicklow *et al.*, 2014). Therefore, the function of Oct4 is likely limited by the insufficient levels of Sox2 and possibly other factors in morulae. Another plausible limiting aspect could be epigenetic constraints in morulae. The epigenetic state and 3-dimensional structure of the chromatin in the morula differ from those in the ICM, which may hinder Oct4 binding (Du *et al.*, 2017; Wang *et al.*, 2018; Wang *et al.*, 2014; Zhang *et al.*, 2016). However, considering that forced expression of Oct4 can sustain cell proliferation and pluripotency in *Sox2*-KO ESCs (Masui *et al.*, 2007), it's plausible that Oct4 and Sox2 may compensate each other in the morula as well. To delve deeper into this possibility,

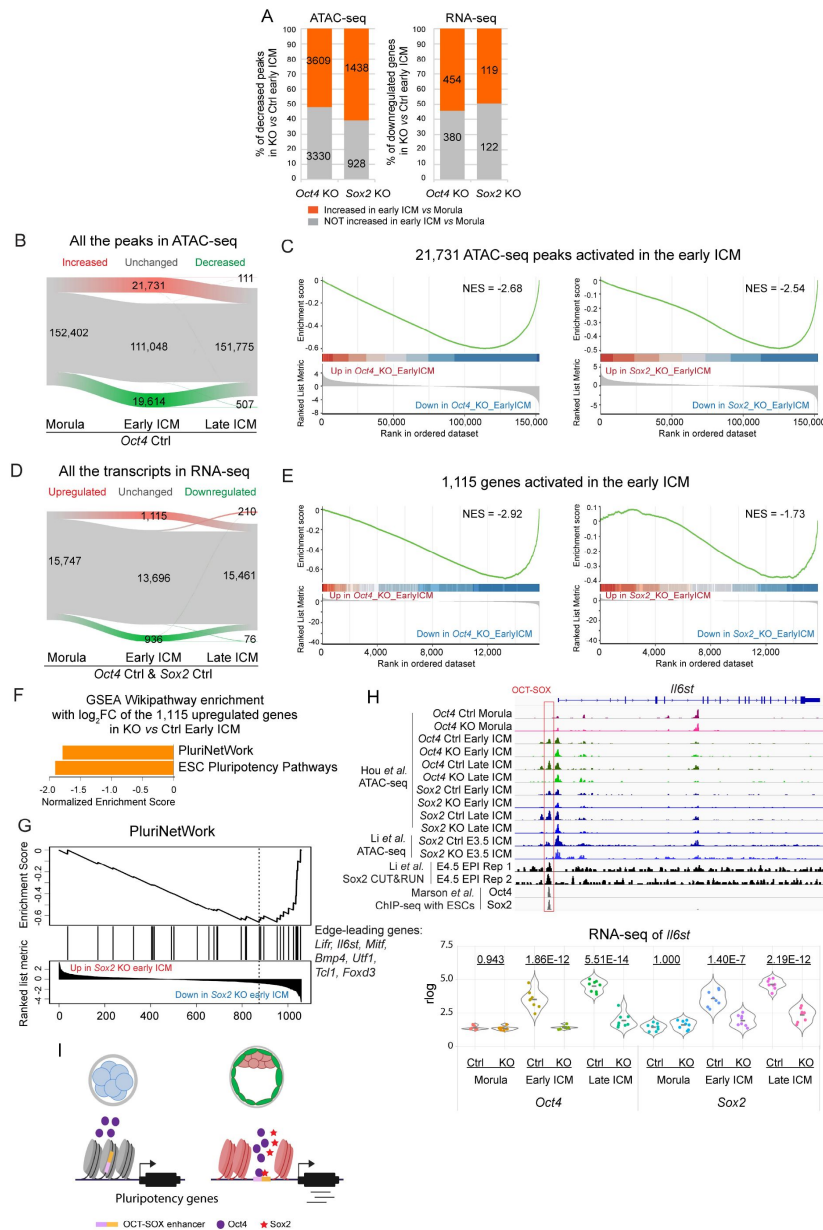


Figure 5.

Oct4 and Sox2 promote the developmental trajectory from the morula to the ICM.

A. Bar graphs show the dynamics of the decreased peaks (left) and genes (right) in *Oct4*- and *Sox2*-KO early ICMs from the morula to the early ICM. **B&D.** Alluvial plots show the dynamics of chromatin accessibility (B) and transcriptome (D) from morula to late ICM in the Ctrl embryos. Green, grey and red lines represent the decreased, unchanged and increased peaks/genes, respectively. In D, only genes significantly up- or down-regulated in both *Oct4* Ctrl and *Sox2* Ctrl embryos were considered as up- or down-regulated genes, while all the rest were considered as unchanged genes. **C&E.** GSEA plots show the enrichment of the 21,731 ATAC-seq peaks (C) and 1,115 genes (E) in *Oct4*-KO and *Sox2*-KO early ICMs. NES, normalized enrichment score. **F.** The bar chart illustrates the GSEA Wikipathway enrichment in WebGestalt. The log₂ foldchange values of the 1,115 upregulated genes (Figure 5D) in *Sox2* KO vs Ctrl early ICM were used in this analysis. FDR≤0.05. FC, foldchange. **G.** GSEA enrichment plot of the term PluriNetWork in Figure 5F. NES, normalized enrichment score. **H.** Upper panel: IGV tracks displaying ATAC-seq and Sox2 CUT&RUN profiles (Li et al, 2023), along with ChIP-seq profiles of Oct4 and Sox2 in ESCs (Marson et al, 2008), centered around the genomic locus of *Il6st*. Red box mark the OCT-SOX enhancer. Lower panel: violin plot showing the rlog values of *Il6st*. The underlined numbers represent the adjusted p values. **I.** Model of the activation of pluripotency-related genes in the early embryos.

double KO embryos would be necessary for further investigation. A limitation of this study is that the whole morula was used for ATAC-seq and RNA-seq. Considering that morulae contain more TE progenitor cells than ICM progenitor cells, the signal from ICM progenitor cells might be diluted at the morula stage.

Oct4 is activated prior to Sox2 in both embryogenesis and reprogramming to iPSCs. In mouse embryos, Oct4 is initially highly expressed in all blastomeres from the 8-cell stage and later becomes restricted to the EPI, whereas Sox2 is repressed by the Hippo pathway in early stages and is specifically expressed in the EPI (Frum *et al.*, 2019; Guo *et al.*, 2010; Wicklow *et al.*, 2014). In reprogramming, the activation of endogenous Sox2 occurs later than the endogenous Oct4 and signifies the final phase of iPSC generation (Buganim *et al.*, 2012). In both scenarios, the expression of Sox2 coincides with the setup of the pluripotent state. Moreover, our data show that Sox2-KO ICMs fail to fully activate the pluripotent program (Figure 3C). Recently, we reported that the loss of Sox2 might contribute to decreased developmental potential of pluripotent cells upon priming (MacCarthy *et al.*, 2024). This suggests that Sox2 plays a critical role in regulating the spatial-temporal onset of a subgroup of pluripotency-related genes.

Loss of Oct4 and Sox2 impairs embryonic pluripotency, as evidenced by the inability of KO embryos to give rise to ESCs or contribute to the embryonic part in chimera assays (Avilion *et al.*, 2003; Nichols *et al.*, 1998; Wu *et al.*, 2013). To date, only a few of pluripotency-related genes have been found downstream of Oct4 and Sox2 in mouse embryos, whereas the majority, including *Nanog*, *Klf4* and *Zfp42* (*Rex1*), are reportedly independent of Oct4 and Sox2 (Frum *et al.*, 2013; Stirparo *et al.*, 2021; Wu *et al.*, 2013). Here, we showed that both Oct4 and Sox2 are essential for establishing the pluripotent chromatin landscape and transcriptome (Figure 3A, 3C, 5F, 5G). We found that a large number of pluripotency-related genes were downregulated in Oct4-KO ICM, contrasting with previous reports which suggested that most pluripotency-related genes were maintained (Frum *et al.*, 2013; Le Bin *et al.*, 2014; Stirparo *et al.*, 2021; Wu *et al.*, 2013). This discrepancy may be due to differences in the experimental design. First, unlike previous study that used the whole blastocysts (Frum *et al.*, 2013), we managed to isolate a pure EPI cell population performing immunosurgery and using short-term inhibition of MEK to reduce the heterogeneity of the ICMs while it did not alter the target enhancers and genes of Oct4 and/or Sox2 (Nichols *et al.*, 2009) (Figure 1-figure supplement 2; Figure 2-figure supplement 3B-3D, 4B-4D). Second, we removed Oct4 exons 2 to 5 (Figure 1A), encompassing the whole POU domain (bipartite DNA-binding domain of Oct4), resulting in a complete ablation of Oct4 function. In contrast, some of the previous studies removed only exon 1 (Frum *et al.*, 2013; Wu *et al.*, 2013), which encodes for N-terminal transactivation domain. The remaining exons 2 to 5 might be translated into a truncated version of Oct4 that could retain partial function. Finally, unlike previous study that used maternal-WT zygotic-KO embryos (Stirparo *et al.*, 2021), we used maternal-zygotic-KO embryos (Figure 1C). The residual levels of maternal Oct4 could have potentially rescued the KO.

To regulate the expression of developmental genes, preimplantation embryos undergo dynamic epigenomic reprogramming, governed by multiple epigenetic modifiers and complexes. For example, *Ezh2* is required for the establishment and maintenance of global H3K27me3 in preimplantation embryos; *Uhrf1* maternal-KO embryos fail to form healthy morula due to misregulation of DNA methylation and histone modifications (Cao *et al.*, 2019; Maenohara *et al.*, 2017; Puschendorf *et al.*, 2008). Our data show that Oct4 and Sox2 activated the components of several epigenetic modifiers, including *Ezh2*, *Uhrf1* and *Sap30* (Figure 2F, 2G; Figure 2-figure supplement 3E; Data supplement 2), and thus may be involved this way in regulating the epigenetic status.

The core regulatory circuitry of pluripotency exhibits remarkable stability in embryos. Expression of the TFs within this circuitry, such as *Oct4*, *Sox2*, *Nanog*, *Klf4* and *Esrrb*, is partially or fully maintained in the Oct4- or Sox2-KO ICMs (Le Bin *et al.*, 2014; Wicklow *et al.*, 2014; Wu *et al.*,

2013) (Figure 3C, 3D). The independent and robust expression of these TFs likely ensures the faithful activation of the pluripotency regulatory circuit, safeguarding successful development.

While the cooperative role of Oct4 and Sox2 on the OCT-SOX enhancers in maintaining or inducing the pluripotent network is widely acknowledged, there are still debates regarding the precise mechanisms underlying their cooperation (Biddle *et al.*, 2019; Chen *et al.*, 2008; Han *et al.*, 2022; Li *et al.*, 2019; Michael *et al.*, 2020; Velychko *et al.*, 2019). In this study, most OCT-SOX peaks are more affected in *Oct4*-KO ICMs than in *Sox2*-KO ICMs, indicating that Oct4 plays a more important role than Sox2 in activating OCT-SOX enhancers. This observation aligns with a previous study showing that the forced expression of Oct4 can rescue pluripotency in *Sox2*-null ESCs (Masui *et al.*, 2007). In addition to their cooperation on OCT-SOX enhancers, recent studies have unveiled independent functions of Oct4 and Sox2 (Friman *et al.*, 2019; Gao *et al.*, 2022). Consistently, we observed that Oct4 regulates a considerable number of peaks and genes independently of Sox2 in the ICMs (Figure 2C, 3A, 3C, 3D). We believe that this outcome is unlikely to be attributed to compensation from other members in the Sox family. Sox1, Sox3, Sox15 and Sox18 are promising candidates as they can either rescue the *Sox2*-KO phenotype in mESCs or replace Sox2 to generate iPSCs in reprogramming (Nakagawa *et al.*, 2008; Niwa *et al.*, 2016). However, our analysis reveals that *Sox1*, *Sox3* and *Sox18* are expressed at extremely low levels in both wildtype and *Sox2*-KO embryos (Data supplement 2), suggesting that they are unlikely to be able to fulfill Sox2's role (Deng *et al.*, 2014). Sox15 exhibits indistinguishable expression levels in all three cell types at the blastocyst stage. Furthermore, *Sox15* KO mice displays normal health and fertility (Lee *et al.*, 2004). Nonetheless, we cannot completely rule out the possibility of functional compensation by Sox15 in *Sox2*-KO ICMs.

Our results highlight the crucial role of Oct4 and Sox2 in establishing the transcriptome and chromatin state in the pluripotent EPI. At the blastocyst stage, Oct4 and Sox2 work together to open the OCT-SOX enhancers and activate the pluripotency-related genes (Figure 5I). The absence of Sox2 and other factors likely limits the function of Oct4 in morulae. However, the upstream factors driving the activation of the core pluripotency regulatory circuitry remain unknown. Further studies are needed to deepen our understanding of the molecular mechanisms governing the embryonic pluripotency program and the early lineage segregation.

Materials and methods

Mice

To generate ESCs carrying a floxed *Oct4* allele (*Oct4^{fllox}*), the construct containing floxed *Oct4* exon 2-5 and a promoter-less FRT-IRES-βgeo-pA cassette was electroporated into germline-competent Acr-EGFP ESCs. Clones were screened for homologous recombination and transiently transfected with an FLP expression vector to remove the FRT cassette. To generate ESCs carrying a mutant *Oct4* allele linked to *mKO2* (*Oct4^{mKO2}*), Acr-EGFP ESCs were targeted with the construct containing a PGK-pacΔtk-P2A-mKO2-pA cassette 3.6 kb upstream of the *Oct4* TSS and a promoter-less FRT-SA-IRES-hph-P2A-Venus-pA cassette in *Oct4* intron 1. To generate ESCs carrying the floxed *Sox2* allele (*Sox2^{fllox}*), the construct containing the 5' loxP site 1.9 kb upstream of *Sox2* TSS and a promoter-less FRT-IRES-Neo-pA cassette followed by the 3' loxP site in the 3' UTR of *Sox2* was electroporated into Acr-EGFP ESCs. Clones were screened for homologous recombination and transiently transfected with an FLP expression vector to remove the FRT cassette. To generate ESCs carrying a conditional allele of *Sox2* linked to *EGFP* (*Sox2^{EGFP}*), Acr-EGFP ESCs were targeted with the construct containing the 5' loxP site 1.9 kb upstream of *Sox2* TSS, and a promoter-less FRT-IRES-Neo-pA cassette, the 3' loxP site, and a PGK-EGFP-pA cassette in the 3' UTR of *Sox2*. Chimeric mice were generated by morula aggregation and heterozygous mice were obtained through germline transmission. The Zp3-Cre transgenic mice were used to generate maternal KO alleles. Animal experiments and husbandry were performed according to the German Animal Welfare guidelines

and approved by the Landesamt für Natur, Umwelt und Verbraucherschutz Nordrhein-Westfalen (State Agency for Nature, Environment and Consumer Protection of North Rhine-Westphalia). The work was funded by the Max Planck Society and the Max Planck Society's White Paper-Project "Animal testing in the Max-Planck- Society".

Embryo collection

All the embryos for RNA-seq and ATAC-seq were collected from female mice superovulated with PMSG and hCG. Early and late morula samples were collected at E2.75 and E3.25, respectively. Early and late ICM samples were collected at E2.5 and cultured in KSOM with MEKi (PD0325901, 1 μ M) until early and late blastocyst stages, respectively.

ICM isolation

Zona pellucida was removed by brief incubation in prewarmed acidic Tyrode's solution. ICM was obtained by immunosurgery. Briefly, blastocysts were incubated in 20% rabbit anti-mouse whole serum (Sigma-Aldrich) in KSOM at 37°C, 5% CO₂ for 15 minutes, followed by three rinses in M2. Afterwards, embryos were incubated in 20% guinea pig complement serum (Sigma-Aldrich) in KSOM at 37°C, 5% CO₂ for 15 minutes, followed by three rinses in M2. In the end, ICM was isolated by repetitive blowing in mouth pipette to remove the debris of dead trophectoderm cells.

Immunofluorescence

Immunofluorescence was performed using the following primary antibodies with dilutions: mouse monoclonal anti-Oct4 (sc-5279, Santa Cruz), 1:1000; goat anti-Sox2 (GT15098, Neuromics), 1:300; rabbit anti-mKO2 (PM051, MBL), 1:1000; rabbit anti-GFP (ab290, Abcam), 1:500; goat anti-Pecam1 (AF3628, R&D biosystems), 1:300; rat monoclonal anti-Nanog (14- 5761-80, Thermo Fisher Scientific), 1:100; goat anti-Sox17 (AF1924, R&D Systems), 1:1000 of 0.2 mg/ml.

ATAC-seq and data analysis

ATAC-seq libraries were prepared as previously described with some modifications. 20 to 40 morulae or ICMs were pooled and lysed in the lysis buffer (10 mM Tris-HCl, pH=7.4; 10 mM NaCl; 3 mM MgCl₂; 0.15% Igepal CA-630; 0.1% Tween-20) for 10 min on ice. We added 0.1% Tween-20 to the lysis buffer as it greatly reduced reads mapped to mitochondrial DNA. Lysed embryos were briefly washed, collected in 2.0 μ l PBS containing 0.1 mg/ml polyvinyl alcohol (PVA) and transferred to 3.0 μ l Tn5 transposome mixture (0.25 μ l Tn5 transposome, 2.5 μ l tagmentation buffer, Illumina, FC-121-1030; 0.25 μ l H₂O). The samples were incubated in 37 °C water bath for 30 min. Tagmentation was stopped by adding 2.0 μ l 175 mM EDTA and incubated at 50 °C for 30 min. Excess EDTA was quenched by 2.0 μ l 160 mM MgCl₂. The libraries were amplified in the following reaction: 9.0 μ l Transposed DNA, 10.0 μ l NEBNext High-Fidelity 2×PCR Master Mix (New England Biolabs, M0541S), 0.25 μ l 100 μ M PCR Index 1, 0.25 μ l 100 μ M PCR Index 2 and 0.5 μ l H₂O. The sequences of index 1 and 2 are in Table supplement 2. The PCR program is as the following: 72 °C for 5 min; 98 °C for 30s; 16 thermocycles at 98 °C for 10 s, 63 °C for 30 s and 72 °C for 1 min; followed by 72 °C 5 min. Amplified libraries were purified twice with 1.2 × AMPure XP beads. The sequencing was performed on the NextSeq 500 system with pair-end 75bp.

Sequence reads were trimmed for adapter sequences using SeqPurge and trimmed reads no shorter than 20 bases were aligned to the mm10 mouse reference genome using Bowtie2 with a maximum fragment size of 2000. Duplicated reads were removed using Picard MarkDuplicates (<https://broadinstitute.github.io/picard/>) and only reads uniquely mapped to the standard chromosomes except chrY and chrM (mitochondrial genome) with mapping quality of at least 30 were used for the following analysis. Reads on the positive and negative strands were shifted +4 bp and -5 bp, respectively. Peak calling was performed using MACS2 with the following parameters: --keep-dup all -g mm --nomodel --shift -50 --extsize 100 -B --SPMR. Peaks that overlap

with the blacklisted regions (<https://sites.google.com/site/anshulkundaje/projects/blacklists> ; <https://sites.google.com/site/atacseqpublic/atac-seq-analysismethods/mitochondrialblacklists-1>) and satellite repeats (RepeatMasker) were removed.

Significant peak summits (q-value ≤ 0.001) from biological replicates were merged for each sample group using BEDTools with a maximum distance between two summits to be merged of 200 bp. Only merged peak summits that overlap with summits from at least two replicates were retained. Reads per million (RPM)-normalized pileup signals in the bedGraph format were converted into bigWig files using the UCSC bedGraphToBigWig tool. The bigWig files for the average signal of biological replicates were generated using the UCSC bigWigMerge tool. The heatmaps of RPM-normalized pileup signals around the TSSs of GENCODE (vM23) protein-coding genes were generated using deepTools.

For count-based analysis of transposon insertion events, merged peak regions were generated by combining the adjacent peak summits of all embryo groups and selecting 200-bp regions around the centers of each merged summit. 5' ends of shifted read alignment on both the positive and negative strands were considered as transposon insertion sites. The number of transposon insertions in each merged peak region were counted for each sample using BEDTools and normalized and transformed to $\log_2$ scale using the rlog function of the Bioconductor package DESeq2 and significantly differential peaks were identified with an adjusted p-value cutoff of 0.05. Peaks located within 100 bp from TSSs of GENCODE (vM23) genes were considered as TSS-proximal peaks. The presence of TF binding motifs in merged peaks was investigated using FIMO with the default cutoff (p-value $< 10^{-4}$).

The following JASPAR motifs were used: MA0036.3 GATA2, MA0139.1 CTCF, MA0141.3 ESRRB, MA0142.1 Pou5f1-Sox2, MA0143.3 Sox2, MA0524.2 TFAP2C, MA0599.1 KLF5, MA0792.1 POU5F1B, MA0800.1 EOMES, MA0808.1 TEAD3 and MA0878.1 CDX1.

Significantly enriched motifs in each peak cluster were identified from using PscanChIP (Zambelli *et al.*, 2013). Functional annotation of gene sets associated with ATAC-seq peaks was performed using GREAT (McLean *et al.*, 2010) with default settings.

RNA-seq and data analysis

Single-embryo RNA-seq was performed as previously described with some modifications (Kurimoto *et al.* 2007, Nakamura *et al.* 2015). 2,493 copies of ERCC RNA Spike-In (Ambion) were added to each single-embryo sample. To remove PCR duplicates, we used R2SP-UMI- d(T)24 primer, instead of V1d(T)24 primer, for reverse transcription (RT). To reduce byproducts derived from the RT primer, the poly(A) tailing reaction was performed only for 1 min at 37°C. cDNA was amplified with V3d(T)24 and R2SP primers and Terra PCR Direct Polymerase (Clontech) for 16 cycles. The cDNA was further amplified for four cycles with NH2- V3d(T)24 and NH2-R2SP primers, and purified thrice with 0.6x AMPure XP beads. The 3' end enriched libraries were constructed using KAPA HyperPlus Library Preparation Kit (VWR International, KK8513) with 6 cycles of PCR. All the oligos used in the RT and library amplification are in Table supplement 3. The sequencing was performed on the NextSeq 500 system with single-end 75bp.

UMI sequence of each read was extracted from FASTQ files for Read 2 using UMI-tools. Poly- A sequences were trimmed from the 3' end of Read 1 using Cutadapt (<https://cutadapt.readthedocs.org/>). The trimmed Read 1 sequences were aligned to the mm10 mouse reference genome using TopHat2 with GENCODE (vM23) transcripts as a transcriptome reference and with default parameters. The reads were also mapped to ERCC reference sequences using Bowtie2 with default parameters. To confirm the genotype of *Oct4*- and *Sox2*-KO embryos, we also mapped the reads to the PGK-pacAtk-P2A-mKO2-hGHpA or PGK-EGFP-hGHpA cassette. The reads mapped to the genome were assigned to GENCODE transcripts using featureCounts. The number of unique transcripts assigned to each GENCODE gene was calculated using TRUmiCount. The transcript counts were

normalized and transformed to $\log_2$ scale using the `rlog` function of the Bioconductor DESeq2 package. Differential expression analysis was performed on genes that were detected in at least half of the samples in at least one stage using DESeq2 and differentially expressed genes were identified with an adjusted p-value <0.05 and $\log_2$ fold change ≥ 1 .

GSEA

Gene set enrichment analysis (GSEA) (Subramanian *et al*, 2005 [\[link\]](#)) was performed to determine whether genes close to the selected ATAC-seq peaks are enriched in genes differentially expressed in *Oct4*- or *Sox2*-KO embryos. The ATAC-seq peaks were annotated to Ensembl protein-coding genes whose TSSs are located within 10 kb of the peak centers using the R package ChIPpeakAnno (Zhu *et al*, 2010 [\[link\]](#)). False discovery rate (FDR) was estimated by gene set permutation tests. Heatmaps were generated from normalized enrichment scores (NESs) for gene sets with FDR ≤ 0.1 .

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Data Availability

ATAC-seq and RNA-seq data have been deposited at GEO under GSE264614 and GSE264615, respectively. Published high-throughput sequencing datasets used in this manuscript are listed as follows: ATAC-seq and CUT&RUN of early embryos, GSE203194; scRNA-seq of early embryos, GSE203194 and GSE159030; Oct4/Sox2/Nanog ChIP of mESCs, GSE11724; H3K27ac ChIP of mESCs, GSE27844. This study includes no data deposited in external repositories.

Author contributions

K.A. and Y.H. designed the study. Y.H. performed the experiments, assembled and interpreted data, and wrote the manuscript. K.A. performed the bioinformatic analysis, interpreted the data and edited the manuscript. G.W. generated the transgenic mice and advised on the project.

Z.N. and S.H. assisted with the experiments. Y.H. and Q.J. contributed to the bioinformatic analysis. S.V. and I.B. advised on the project and edited the manuscript. H.R.S. supervised the project and edited the manuscript.

Figure supplements

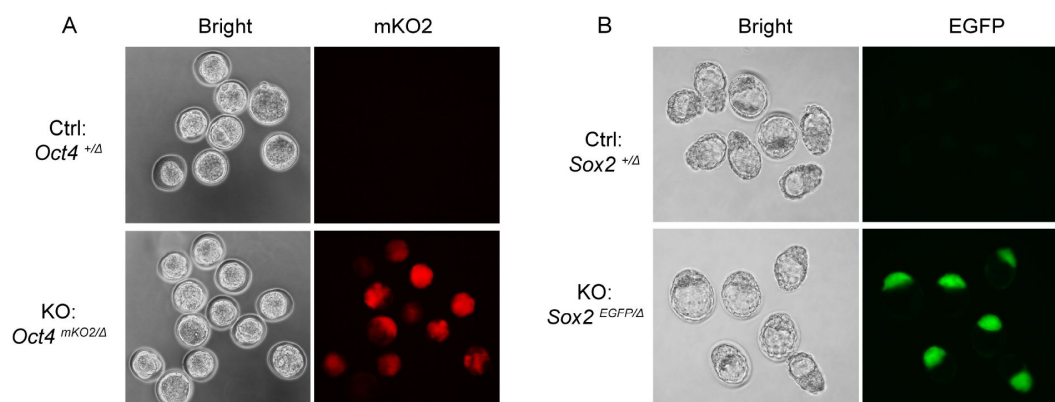


Figure supplement 1.

Validation of the transgenic embryos (related to Figure 1 [↗](#)).

A. The mKO2-labeled *Oct4*-KO embryos were identified under fluorescent microscopy at E2.5+1 day. **B.** The EGFP-labeled *Sox2*-KO embryos were identified under fluorescent microscopy at E2.5+2 days.

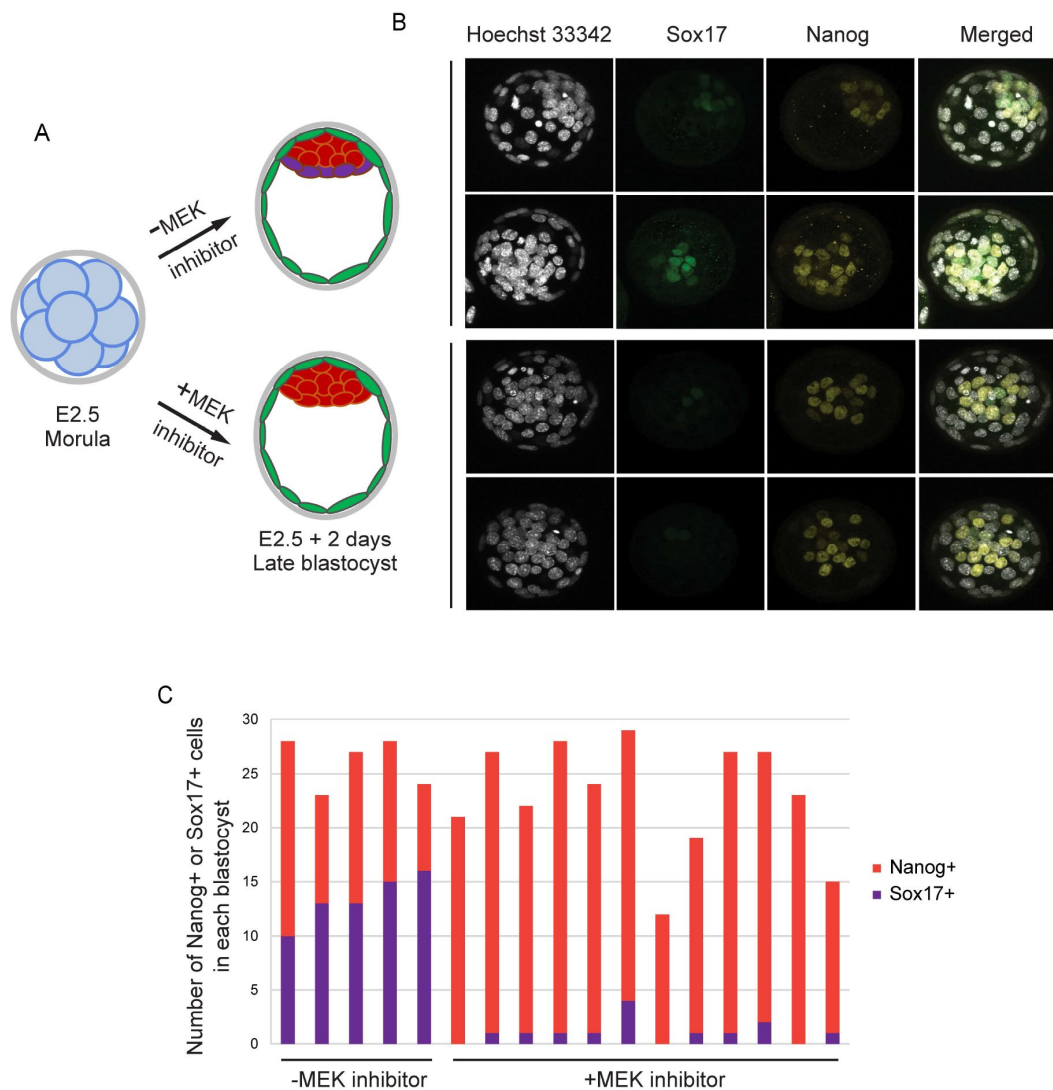


Figure supplement 2.

Validation of the effect of MEK inhibitor (related to Figure 1 [↗](#)).

A. The scheme of MEKi (PD0325901, 1 μ M) treatment. **B.** Immunostaining of Sox17 and Nanog in the late blastocysts treated with or without MEKi. **C.** The number of Sox17+ or Nanog+ cells in each blastocyst.

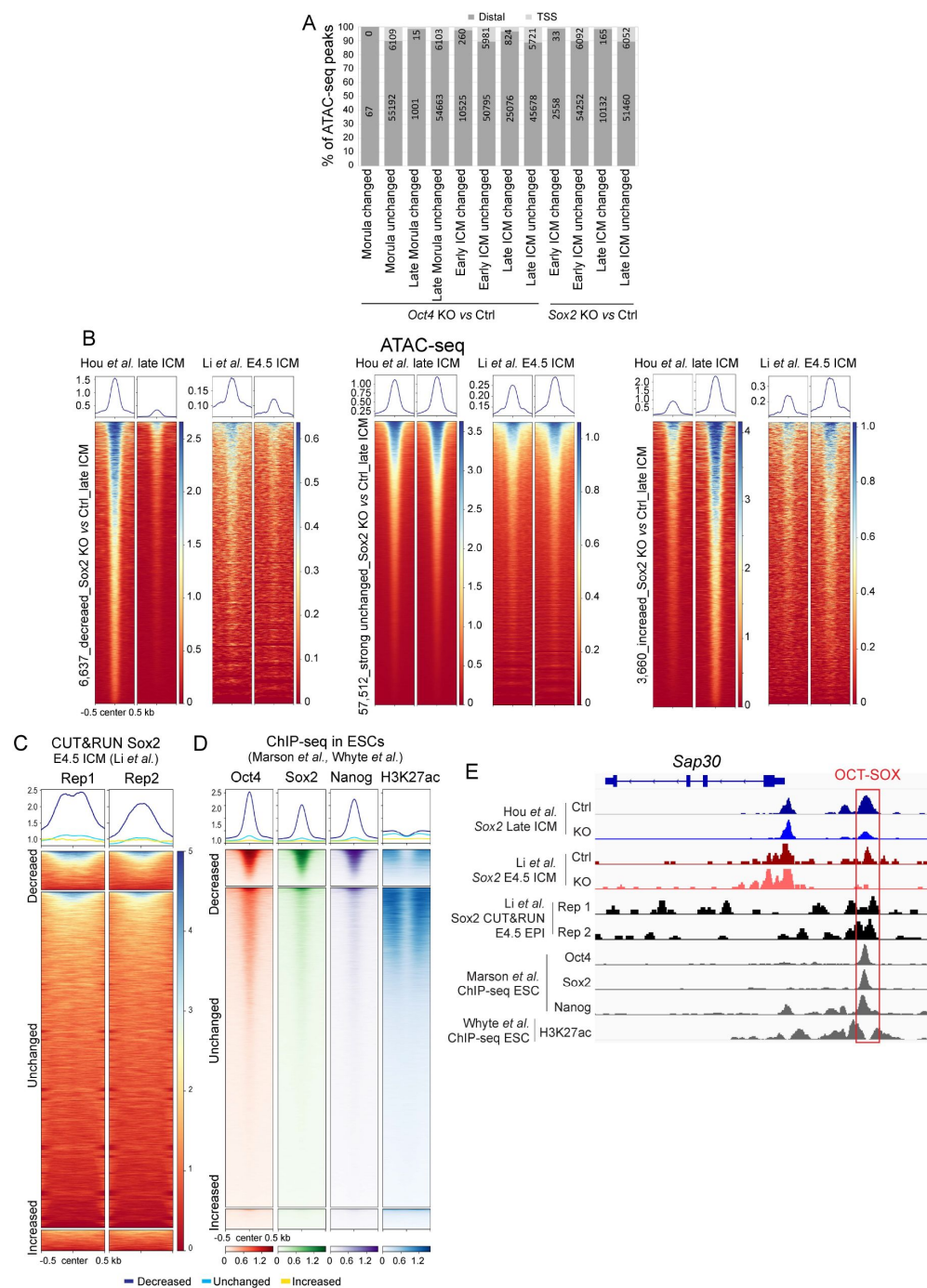


Figure supplement 3.

Quality check of the low-input ATAC-seq (related to Figure 2).

A. The location of ATAC-seq peaks relative to TSSs. The number in each bar represents the number of peaks in each group. **B&C.** ATAC-seq (B) and Sox2 CUT&RUN (C) profiles (Li *et al.*, 2023) over the decreased, unchanged and increased ATAC-seq peaks identified in our Sox2-KO late ICMs. **D.** ChIP-seq profiles of Oct4, Sox2, Nanog and H3K27ac in ESCs (Marson *et al.*, 2008) over the decreased, unchanged and increased ATAC-seq peaks in our Sox2-KO late ICMs. To exclude spurious peaks, only strong unchanged peaks were used in the analysis from C to E. **E.** IGV tracks displaying ATAC-seq and Sox2 CUT&RUN profiles (Li *et al.*, 2023) in late ICMs, along with ChIP-seq profiles of Oct4, Sox2, Nanog and H3K27ac in ESCs (Marson *et al.*, 2008; Whyte *et al.*, 2012), centered around the genomic locus of *Sap30*. Arrows mark the decreased ATAC-seq peaks.

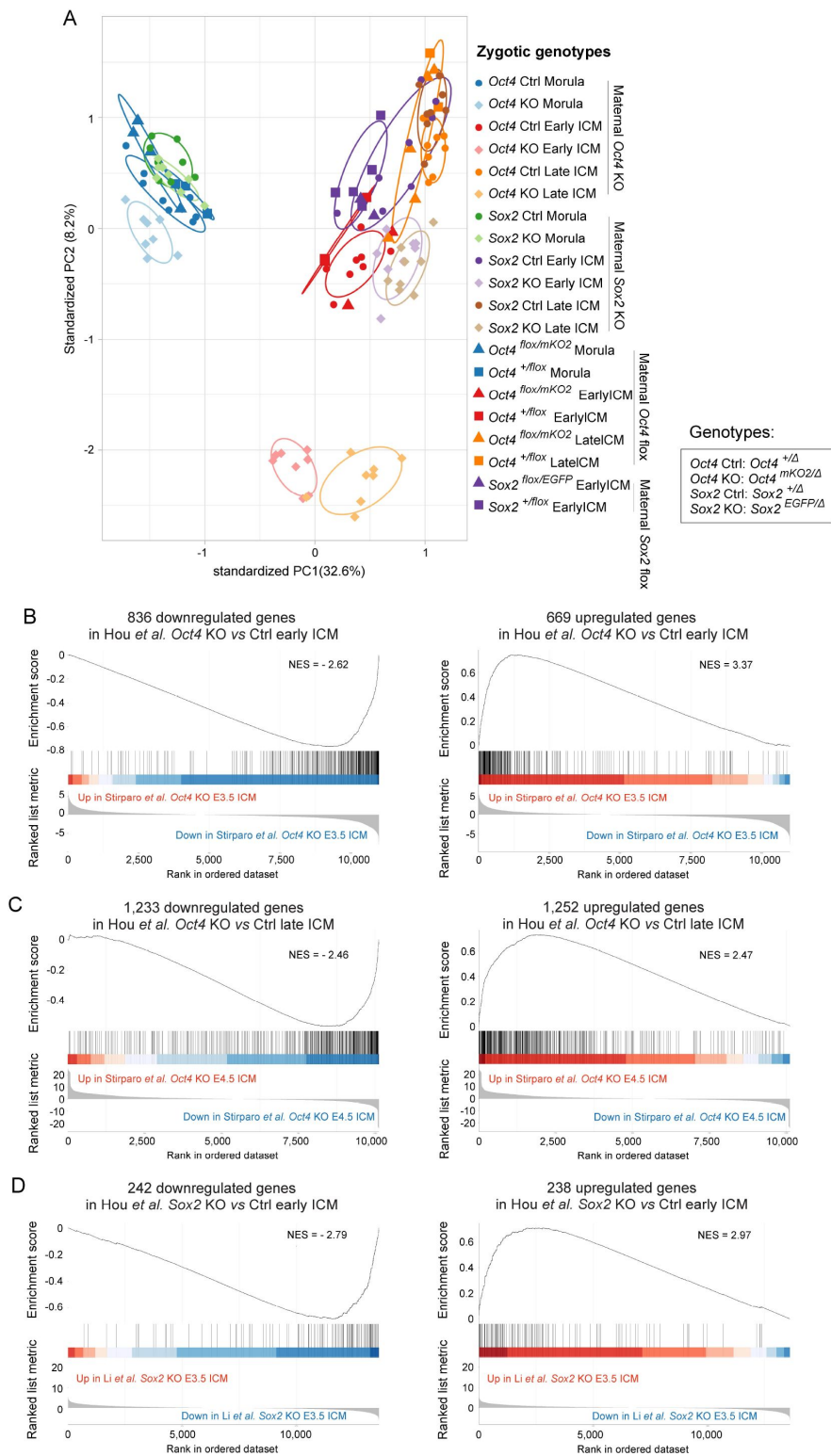


Figure supplement 4.

Quality check of the low-input RNA-seq (related to Figure 2).

A. PCA plot illustrating the distribution of RNA-seq samples with the indicated stages and genotypes. **B-D.** GSEA plots of down- and up-regulated genes identified in our Oct4-KO early ICM (B), Oct4-KO late ICM (C) and Sox2-KO early ICM (D) in the scRNA-seq dataset of ICM samples without MEKi (Stirparo *et al.*, 2021; Li *et al.*, 2023). NES, normalized enrichment score.

Figure supplement 5.

The correlation between chromatin accessibility and the transcription activity of surrounding genes (related to Figure 2).

Heatmap of GSEA enrichment *Oct4*-KO (A) and *Sox2*-KO (B) samples. The analysis utilized genes with TSS falling within a 10 kb proximity to the decreased or increased ATAC-seq peaks. The number in parentheses denotes the count of genes with TSS located within 10 kb of the significantly differential peaks in each module. NES, normalized enrichment score.

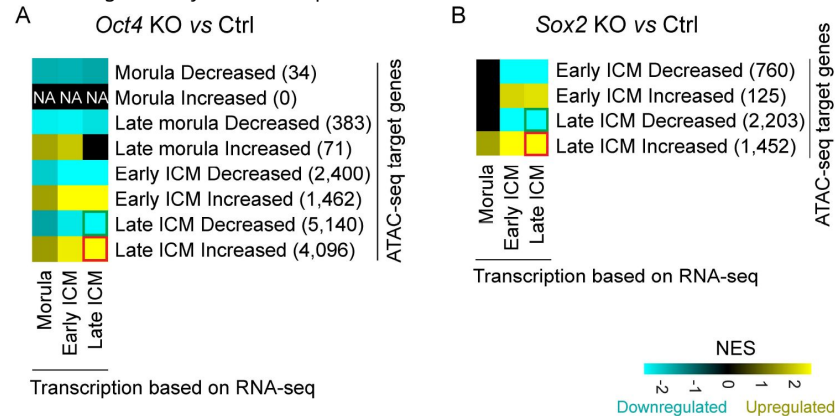
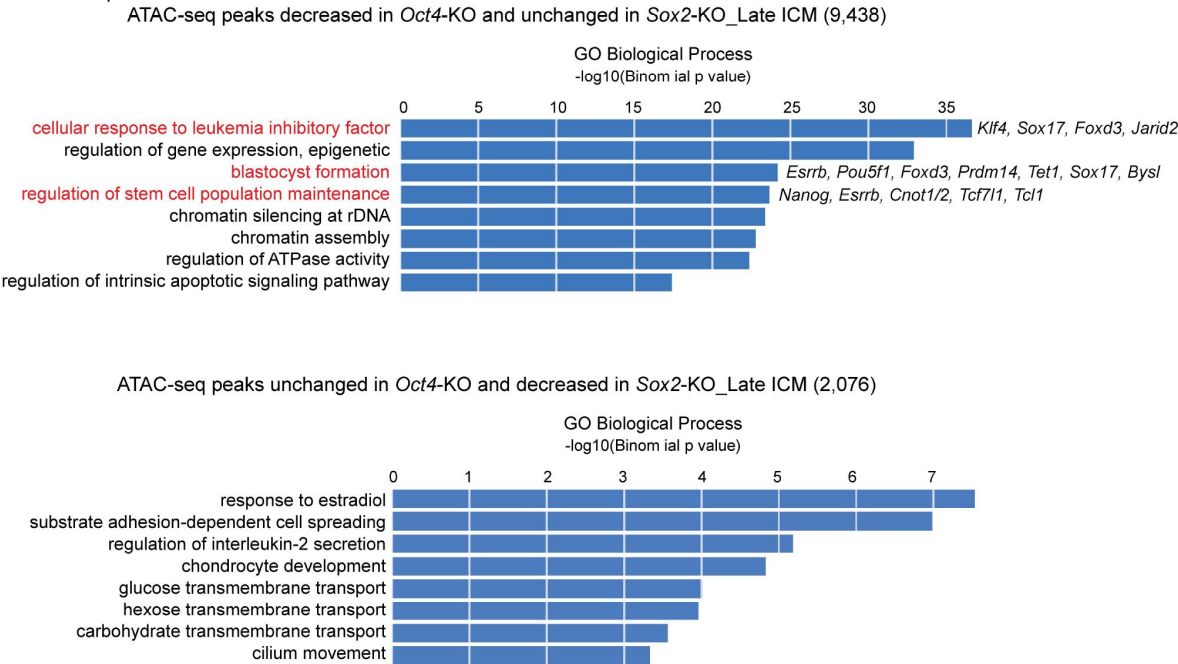


Figure supplement 6.

GREAT Biological Process enrichment analysis of ATAC-seq peaks specifically decreased in the *Oct4*-KO or *Sox2*-KO late ICMs (related to Figure 3).

Terms highlighted in red are related to the pluripotency and preimplantation embryonic development. Example genes of associated peaks are listed.



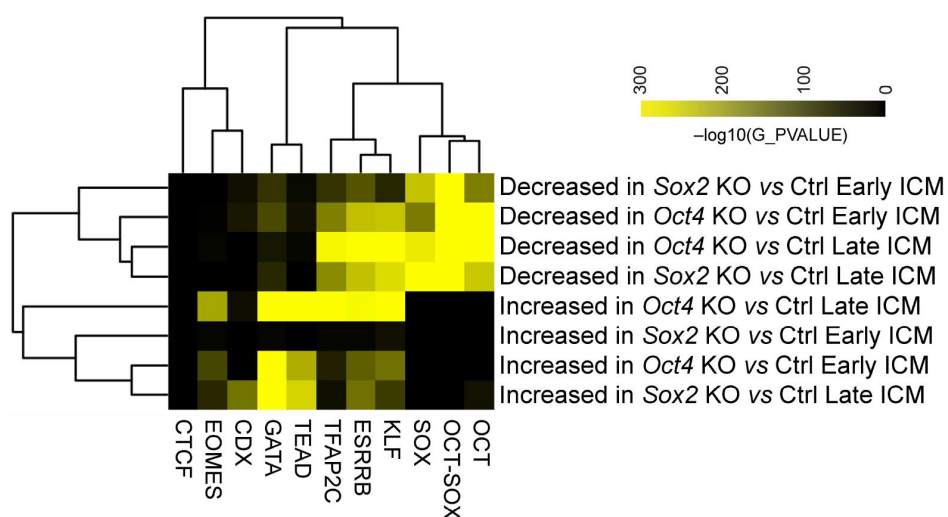


Figure supplement 7.

Motif enrichment analysis of significantly changed ATAC-seq peaks that are distal to TSSs (related to Figure 4 [↗](#)).

The values are $-\log_{10}(\text{p-value})$.

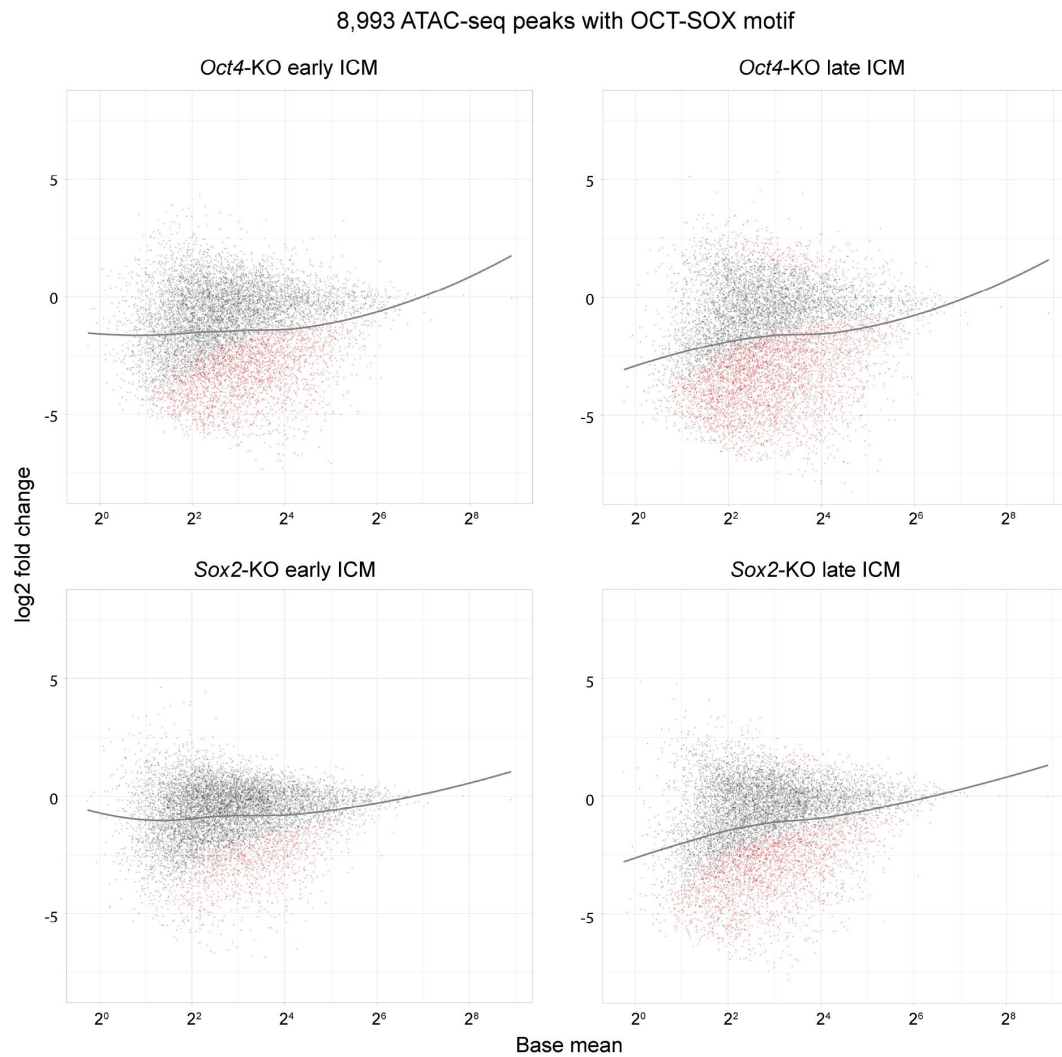


Figure supplement 8.

MA plots show the log₂ fold change of the 8,993 OCT-SOX peaks in *Oct4*- and *Sox2*-KO ICMS (related to Figure 4 [↗](#)).

Black and red dots represent the unchanged and significantly changed peaks, respectively. Grey lines are the regression curves of the 8,993 OCT-SOX peaks.

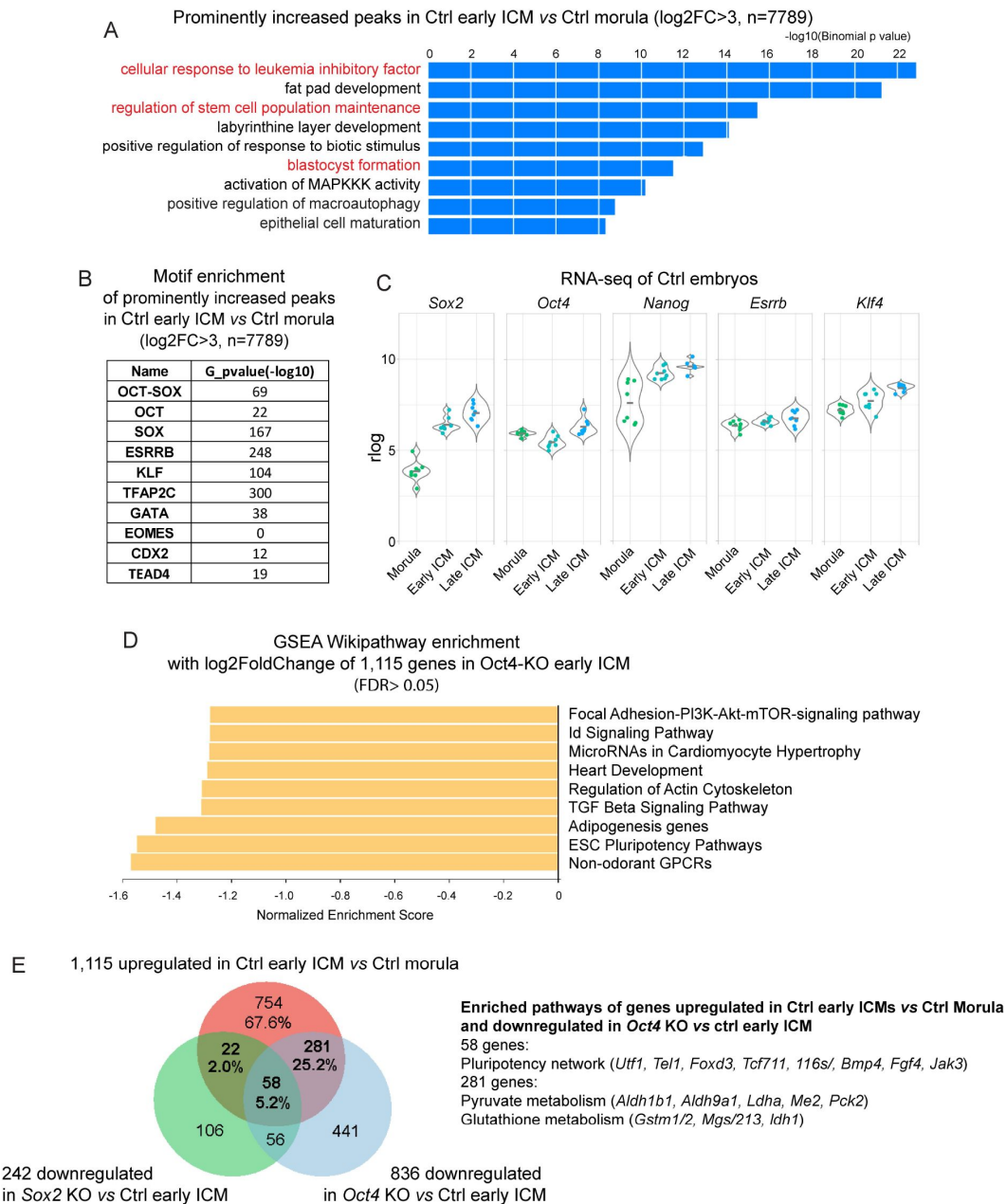


Figure supplement 9.

The dynamics of chromatin accessibility and transcriptome from morula to ICM (related to Figure 5 [↗](#)).

A GREAT analysis was conducted on the most prominently elevated ATAC-seq peaks in *Ctrl* early ICM vs *Ctrl* morula. A $\log_2$ fold change cutoff of >3 was applied due to the limits of GREAT analysis with fewer than 10,000 peaks. **B**. Motif enrichment analysis with PScan of the increased ATAC-seq peaks in *Ctrl* early ICM vs *Ctrl* morula. Cutoff, $\log_2$ fold change >3 . **C**. Expression of *Sox2*, *Oct4*, *Nanog*, *Esrrb* and *Klf4* in morulae and ICMs (related to Figure 5 [↗](#)). **D**. Bar chart illustrating the GSEA Wikipathway enrichment in WebGestalt. The $\log_2$ foldchange values of the 1,115 upregulated genes (Figure 5D [↗](#)) in *Oct4* KO vs *Ctrl* early ICM were used in this analysis. $FDR > 0.05$. **E**. Venn diagram showing the overlap between genes upregulated in early ICMs and those downregulated in *Oct4*-KO or *Sox2*-KO early ICMs. Pathway over-representation analysis was conducted on the genes upregulated in *Ctrl* early ICMs vs *Ctrl* Morula and downregulated in *Oct4* KO vs *Ctrl* early ICM.

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Reviewer #1 (Public review):

Summary:

Numerous mechanism and structural studies reported the cooperative role of Oct4 and Sox2 during the establishment of pluripotency during reprogramming. Due to the difficulty in sample collection and RNA-seq with low-number cells, the precise mechanisms remain in early embryos. This manuscript reported the role of OCT4 and SOX2 in mouse early embryos using knockout models with low-input ATAC-seq and RNA-seq. Compared to the control, chromatin accessibility and transcriptome were affected when Oct4 and Sox2 were deleted in early ICM. Specifically, decreased ATAC-seq peaks showed enrichment of Motifs of TF such as OCT, SOX, and OCT-SOX, indicating their importance during early development. Moreover, by deep analysis of ATAC-seq and RNA-seq data, they found Oct4 and Sox2 target enhancer to activate their downstream genes. In addition, they also uncovered the role of OS during development from the morula to ICM, which provided the scientific community with a more comprehensive understanding.

Strengths:

On the whole, the manuscript is innovative, and the conclusions of this paper are mostly well supported by data, however, there are some issues that need to be addressed.

Weaknesses:

Major Points:

- (1) In Figure 1, a more detailed description of the knockout strategy should be provided to clarify itself. The knockout strategy in Fig1 is somewhat obscure, such as how is OCT4 inactivated in Oct4mKO2 heterozygotes. As shown in Figure 1, the exon of OCT4 is not deleted, and its promoter is not destroyed. Therefore, how does OCT4 inactivate to form heterozygotes?
- (2) Is ZP 3-Cre expressed in the zygotes? Is there any residual protein?
- (3) What motifs are enriched in the rising ATAC-seq peaks after knocking out of OCT4 and SOX2?
- (4) The ordinate of Fig4c is lost.

(5) Signals of H3K4me1, H3K27ac, and so on are usually used to define enhancers, and the loci of enhancers vary greatly in different cells. In the manuscript, the authors defined ATAC-seq peaks far from the TSS as enhancers. The definition in this manuscript is not strictly an enhancer.

(6) If Oct4 and Sox2 truly activate *sap 30* and *Uhrf 1*, what effect does interfering with both genes have on gene expression and chromatin accessibility ?

<https://doi.org/10.7554/eLife.100735.1.sa2>

Reviewer #2 (Public review):

In this manuscript, Hou et al. investigate the interplay between OCT4 and SOX2 in driving the pluripotent state during early embryonic lineage development. Using knockout (KO) embryos, the authors specifically analyze the transcriptome and chromatin state within the ICM-to-EPI developmental trajectory. They emphasize the critical role of OCT4 and the supportive function of SOX2, along with other factors, in promoting embryonic fate. Although the paper presents high-quality data, several key claims are not well-supported, and direct evidence is generally lacking.

Major Points:

(1) Although the authors claim that both maternal KO and maternal KO/zygotic hetero KO mice develop normally, the molecular changes in these groups appear overestimated. A wildtype control is recommended for a more robust comparison.

(2) The authors assert that OCT4 and SOX2 activate the pluripotent network via the OCT-SOX enhancer. However, the definition of this enhancer is based solely on proximity to TSSs, which is a rough approximation. Canonical enhancers are typically located in intronic and intergenic regions and marked by H3K4me1 or H3K27ac. Re-analyzing enhancer regions with these standards could be beneficial. Additionally, the definitions of "close to" or "near" in lines 183-184 are unclear and not defined in the legends or methods.

(3) There is no evidence that the decreased peaks/enhancers could be the direct targets of Oct4 and Sox2 throughout this manuscript. Figures 2 and 4 show only minimal peak annotations related to OCT and SOX motifs, and there is a lack of chromatin IP data. Therefore, claims about direct targets are not substantiated and should be appropriately revised.

(4) Lines 143-146 lack direct data to support the claim. Actually, the main difference in cluster I, 11 and 3, 8, 14 is whether the peak contains OCT-SOX motif. However, the reviewer cannot get any information of peaks activated by OCT4 rather than SOX2 in cluster I, 11.

Minor Points:

(1) Lines 153-159: The figure panel does not show obvious enrichment of SOX2 signals or significant differences in H3K27ac signals across clusters, thus not supporting the claim.

(2) Lines 189-190: The term "identify" is overstated for the integrative analysis of RNA-seq and ATAC-seq, which typically helps infer TF targets rather than definitively identifying them.

(3) The Discussion is lengthy and should be condensed.

<https://doi.org/10.7554/eLife.100735.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review)

Summary:

Numerous mechanism and structural studies reported the cooperative role of Oct4 and Sox2 during the establishment of pluripotency during reprogramming. Due to the difficulty in sample collection and RNA-seq with low-number cells, the precise mechanisms remain in early embryos. This manuscript reported the role of OCT4 and SOX2 in mouse early embryos using knockout models with low-input ATAC-seq and RNA-seq. Compared to the control, chromatin accessibility and transcriptome were affected when Oct4 and Sox2 were deleted in early ICM. Specifically, decreased ATAC-seq peaks showed enrichment of Motifs of TF such as OCT, SOX, and OCT-SOX, indicating their importance during early development. Moreover, by deep analysis of ATAC-seq and RNA-seq data, they found Oct4 and Sox2 target enhancer to activate their downstream genes. In addition, they also uncovered the role of OS during development from the morula to ICM, which provided the scientific community with a more comprehensive understanding.

Strengths:

On the whole, the manuscript is innovative, and the conclusions of this paper are mostly well supported by data, however, there are some issues that need to be addressed.

Weaknesses:

Major Points:

(1) In Figure 1, a more detailed description of the knockout strategy should be provided to clarify itself. The knockout strategy in Fig1 is somewhat obscure, such as how is OCT4 inactivated in Oct4mKO2 heterozygotes. As shown in Figure 1, the exon of OCT4 is not deleted, and its promoter is not destroyed. Therefore, how does OCT4 inactivate to form heterozygotes?

Thank you for your kind suggestions. We will add a detailed description of the knockout strategy in the legends for Figure 1A and 1B, as shown below:

Figure 1A. Schemes of mKO2-labeled Oct4 KO (Oct4mKO2) and Oct4 flox alleles. In the Oct4mKO2 allele, a PGK-pacΔtk-P2A-mKO2-pA cassette was inserted 3.6 kb upstream of the Oct4 transcription start site (TSS) and a promoter-less FRT-SA-IRES-hph-P2A-Venus-pA cassette was inserted into Oct4 intron 1. The inclusion of a stop codon followed by three sets of polyadenylation signal sequences (pA) after the Venus cassette ensures both transcriptional and translational termination, effectively blocking the expression of Oct4 exons 2–5.

Figure 1B. Schemes of EGFP-labeled Sox2 KO (Sox2EGFP) and Sox2 flox alleles. In the Sox2EGFP allele, the 5' untranslated region (UTR), coding sequence and a portion of the 3' UTR of Sox2 were deleted and replaced with a PGK-EGFP-pA cassette. Notably, 1,023 bp of the Sox2 3'UTR remain intact.

(2) Is ZP 3-Cre expressed in the zygotes? Is there any residual protein?

Thank you for the question. While we have not directly tested for ZP3-Cre expression in zygotes, the published transcriptome and proteomics data shows that ZP3 is present at both

the transcriptional and protein levels in wild-type zygotes (Deng et al., *Science*, 2014; Gao et al., *Cell Reports*, 2017). This suggests that ZP3-Cre could potentially be expressed in zygotes as well.

(3) What motifs are enriched in the rising ATAC-seq peaks after knocking out of OCT4 and SOX2?

Thank you for the question. The enriched motifs in the rising ATAC-seq peaking in *Oct4* KO and *Sox2* KO ICMs are the GATA, TEAD, EOMES and KLF motifs, as shown in Figure 4A and Figure supplement 7.

(4) The ordinate of Fig4c is lost.

Thank you for the question. The y-axis is average normalized signals (reads per million-normalized pileup signals). We will add it in the revised version.

(5) Signals of H3K4me1, H3K27ac, and so on are usually used to define enhancers, and the loci of enhancers vary greatly in different cells. In the manuscript, the authors defined ATAC-seq peaks far from the TSS as enhancers. The definition in this manuscript is not strictly an enhancer.

Thank you for this insightful comment. We will search for and analyze published omics data on H3K4me1 and H3K27ac in early embryos or mouse embryonic stem cells to conduct this analysis.

(6) If *Oct4* and *Sox2* truly activate *sap 30* and *Uhrf 1*, what effect does interfering with both genes have on gene expression and chromatin accessibility?

Thank you for the interesting question. Unfortunately, we have not conducted this specific experiment, so we do not have direct results. However, Sap30 is a key component of the mSin3A corepressor complex, while Uhrf1 regulates the establishment and maintenance of DNA methylation. Both proteins are known to function as repressors. Therefore, we hypothesize that interfering with these two genes could alleviate repression of some genes, such as trophectoderm markers, similar to what we have observed in *Oct4* KO and *Sox2* KO ICMs.

Reviewer #2 (Public review):

In this manuscript, Hou et al. investigate the interplay between OCT4 and SOX2 in driving the pluripotent state during early embryonic lineage development. Using knockout (KO) embryos, the authors specifically analyze the transcriptome and chromatin state within the ICM-to-EPI developmental trajectory. They emphasize the critical role of OCT4 and the supportive function of SOX2, along with other factors, in promoting embryonic fate. Although the paper presents high-quality data, several key claims are not well-supported, and direct evidence is generally lacking.

Major Points:

(1) Although the authors claim that both maternal KO and maternal KO/zygotic hetero KO mice develop normally, the molecular changes in these groups appear overestimated. A wildtype control is recommended for a more robust comparison.

Thank you for your valuable feedback. However, I'm unclear on what is meant by "the molecular changes in these groups appear overestimated." Could the reviewer kindly provide

more details or clarify which specific aspects of the molecular changes they are referring to? This would help us better address the concern.

(2) The authors assert that OCT4 and SOX2 activate the pluripotent network via the OCT-SOX enhancer. However, the definition of this enhancer is based solely on proximity to TSSs, which is a rough approximation. Canonical enhancers are typically located in intronic and intergenic regions and marked by H3K4me1 or H3K27ac. Re-analyzing enhancer regions with these standards could be beneficial. Additionally, the definitions of "close to" or "near" in lines 183-184 are unclear and not defined in the legends or methods.

Thank you for this insightful comment. We will search for and analyze published omics data on H3K4me1 and H3K27ac in early embryos or mouse embryonic stem cells to address the concern of "enhancer".

The definition of "close to" or "near" in lines 183-184 is in the legend of Figure 2E and methods. In the GSEA analysis, Ensembl protein-coding genes with TSSs located within 10 kb of ATAC-seq peak centers were included.

(3) There is no evidence that the decreased peaks/enhancers could be the direct targets of Oct4 and Sox2 throughout this manuscript. Figures 2 and 4 show only minimal peak annotations related to OCT and SOX motifs, and there is a lack of chromatin IP data. Therefore, claims about direct targets are not substantiated and should be appropriately revised.

Thank you for the comment. In Figure Supplement 3C, we analyzed published Sox2 CUT&RUN data from E4.5 ICMs (Li et al., Science, 2023), which demonstrates that the reduced ATAC-seq peaks in our Sox2 KO ICMs are enriched with Sox2 CUT&RUN signals. This data suggests that decreased peaks/enhancers could be the direct targets of Sox2. Unfortunately, we did not find similar published data for Oct4 in embryos.

(4) Lines 143-146 lack direct data to support the claim. Actually, the main difference in cluster 1, 11 and 3, 8, 14 is whether the peak contains OCT-SOX motif. However, the reviewer cannot get any information of peaks activated by OCT4 rather than SOX2 in cluster 1, 11.

Thank you for the comment. As the reviewer pointed out, we agree that clusters 3, 8, 14 is more enriched with OCT-SOX motifs than clusters 1/11. However, this is consistent with our observation that the accessibility of peaks in clusters 1 and 11 mainly relies on Oct4, while the accessibility of clusters 3, 8, 14 relies on both Oct4 and Sox2. Probably the word "activate" is not accurate. We will rearrange the texts as below:

"Notably, compared to the peaks dependent on Oct4 but not Sox2 (Figure 2B, clusters 1 and 11), those reliant on both Oct4 and Sox2 show greater enrichment of the OCT-SOX motif (Figure 2B, clusters 3, 8 and 14). The former group tended to be already open in the morula, while the latter group became open in the ICM."

Minor Points:

(1) Lines 153-159: The figure panel does not show obvious enrichment of SOX2 signals or significant differences in H3K27ac signals across clusters, thus not supporting the claim.

Thank you for the comments.

Line 153-159 reference two datasets: Figure supplement 3C and 3D.

In Figure supplement 3C, the average plots above the heatmaps show that the decreased ATAC-seq peaks exhibited higher enrichment with Sox2 CUT&RUN signals compared to the increased or unchanged peaks.

Regarding Figure supplement 3D, we agree that the H3K27ac signal is only slightly more enriched on the decreased peaks than the unchanged peaks. However, it's important to note that only the top 57,512 strongest of the 142,096 unchanged peaks were included in the analysis. We excluded the weaker unchanged peaks because they are less informative. but if included, they could reduce the average H3K27ac signal for the unchanged peaks.

| (2) Lines 189-190: The term "identify" is overstated for the integrative analysis of RNA-seq and ATAC-seq, which typically helps infer TF targets rather than definitively identifying them.

Thank you for the suggestion. We will replace "identify" with "infer". The revised version is as below:

"In addition, integration of the ATAC-seq and RNA-seq data allowed us to **infer** previously unknown targets of Oct4 and Sox2, such as Sap30 and Uhrf1, which are essential for somatic cell reprogramming and embryonic development."

| (3) The Discussion is lengthy and should be condensed.

Thank you for the suggestion. We will shorten it.

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